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DESIGN AND DEVELOPMENT OF AN AUTONOMOUS CAGED DRONE FOR LEAK DETECTION IN HVAC DUCTS

by

Ahmed Khalil

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American University of Sharjah
College of Engineering
in Partial Fulfillment
of the Requirements
for the Degree of

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Declaration of Authorship

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Dedication

To my lovely parents, sister, brothers and fiance´ . . .

Abstract

Interest in unmanned aerial vehicles (UAVs) has increased over the past years mainly due to the benefits that it proposes to achieve. Their ability to perform tasks accurately and efficiently in dangerous conditions makes them an optimal choice in various indoor applications. However, the problem of increased safety and precise control arises as these drones have high rotatory components. To avoid the rotary components being exposed, a cage type structure can be used which helps dampen an impact on the drone. For precise control, advanced adaptive control techniques can be used to achieve a stable vehicle path. In this thesis a hybrid rolling and flying multi-rotor drone is developed for inspection and leak detection in Heating, Ventilation, and Air Conditioning (HVAC) systems ducts through a thermal camera. A multi-rotor drone with a passive cage is designed, manufactured and controlled autonomously using Direct Model Reference Adaptive Control (DMRAC). DMRAC is coupled with a structure of PID controllers. Four methods are used and compared to overcome the drifting gain problem; dead-zone modification, sigma modification, epsilon modification and projection based modification. Additionally, more parameters are studied inside these modifications. The simulations show that all methods complete the desired route with some differences, such as gains drifting and error. The experimental multi-rotor platform is equipped with a 2D LiDAR sensor that provides a scan of the distance surrounding the multi-rotor drone in a 2D plane. An IMU sensor is used to project these distance measurements to a plane parallel to the ground along with a thermal camera for detecting leaks. The projected readings from the LiDAR is compared with the ground truth by using an Optitrack motion capture system. The platform is validated with PID and DMRAC control and the output result is compared amongst themselves, a thermal leak is detected and localized autonomously by the proposed system during real time validation.

Keywords: ***Quadrotor, Caged Drone, 3D printing, UAV, PID, Adaptive Control, Gain modifications, DMRAC, Leak Detection, HVAC, SLAM, Localization, Optitrack, Thermal Imaging***

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Chapter 1.Introduction

In the past two decades, research on unmanned aerial vehicles (UAVs) and their applications have increased rapidly. They did not only develop in terms of accuracy and performance but have also increased in efficiency regarding any particular application. Quad-rotors are a subset of UAVs that have gained significant attention when compared to the rest of the UAVs. Their main attribute is that they have four rotors in a specific configuration and are either autonomously controlled or piloted remotely. As they became more popular, different variations on the design and the number of motors used on the drone also emerged such as single rotor, fixed wing, octa-copter and the octa-quad rotor. This not only helped in increasing the load capacity of the vehicle but also made it more stable to control and maneuver accurately. This multiplied its applications. It became possible to not only use it for outdoor applications but also for indoor ones. Various indoor tasks could be achieved efficiently which opened a whole new area of research on such vehicles. This includes optimizing the battery life of the vehicle as well as precise control of it. As indoor applications usually include the vehicle to maneuver in a tight spot, major damage can be occurred to itself and the surroundings if the control of the drone fails. This eventually opened the field for 'caged' drones that have a protective cage to keep the rotating parts inside itself. This allows the drone a bit more flexibility on the control as the danger of hitting an obstacle is significantly reduced. Moreover, the cage can be used as a means to travel along the ground where 3-dimensional maneuver is not required; hence saving battery life.

1.1. Thesis Objectives

Many issues arise when using a vehicle that is based solely on UAV systems in compact areas. Increased battery drainage during locomotion and failure of precise control are two main challenges to face when using UAVs in a confined space [10]. Moreover, a GPS-denied area would also hinder personal control of the vehicle and add more constraints to the application in hand. A small wheeled robot with a distance sensor can be used to localize and navigate effectively indoors in small areas. However, this looses the possibility of navigation in the 3rd dimension. A hybrid system that is able to 'roll'

on the ground while still being able to fly is proposed in this thesis. Some applications require autonomous navigation of the vehicle while still being in a GPS-denied area is also addressed. Suitable sensor comparison and selection is done so that the vehicle is able to navigate autonomously. Emphasis is required in the control of the drone and hence simulation is done on the drone model using Direct Model Reference Adaptive Control. Different modifications in the control law is also simulated to help combat gain drifts due to nonlinearities.

To test the simulation, a drone was also designed and printed using 3D printing. This includes the cage around the drone to test the rolling mechanism. Control laws are also ran on the hardware and are compared amongst themselves. An Optitrack was used to test the results from the LiDAR and were compared by graphing an error histogram. The drone was then run autonomously to detect a hypothetical leak on an HVAC duct. Developing the concept further, a 2D map of the environment was also generated using the selected sensors using SLAM from Google Cartographer.

1.2. Thesis Organization

The thesis chapters are organized as follows:

- Chapter 1 is an introduction on the thesis's main idea and the thesis objective is also stated. It also includes a general literature review on the related designs in the field.
- Chapter 2 describes the different sensors available and compares some of the similar ones. It also goes over the electronic component selection for the drone.
- Chapter 3 covers the main design and 3D printing stage of the drone with the cage. The overall setup with all the electronics on-board is also presented in this chapter.
- Chapter 4 presents the simulation of the Direct Model Reference adaptive control on the drone with different modifications on the adaptive laws.
- Chapter 5 explains how the control was implemented on the hardware. It also compares the LiDAR data with the ground truth (Optitrack). Thermal leak is then detected autonomously by using a thermal camera.

- Chapter 6 goes over SLAM and Google Cartographer. It presents some results generated by the drone non-autonomously.

1.3. Background and Literature Review

1.3.1 Existing hybrid drones

The trend for combining ground locomotion with aerial motion is increasing as more of its advantages are noticed. Earlier stages included the concept of using fixed wings along with a separate motion mechanism for ground movement. The Morphing Micro Air and Land Vehicle [11] is one of the most famous examples that was developed in the early 2000s. The design was inspired by the biological features of an insect and had wings and wheeled legs for motion. However, difficulties were faced when controlling the vehicle and it had inefficient transition between the two modes of movement [11]. These complications could be linked to the idea that fixed wings were used. To avoid this, rotary flight technology was then introduced to overcome these issues [12]. [13] were among the first to use propellers for flight motion while also having wheels on the vehicle for ground movement. Several others followed the same concept with each of them having its own unique feature. The DUCK robot [14] uses the same configuration of the actuators that are used in a quadcopter. It has passive-dynamic legs that are able to move on the ground without the need of any extra actuator. Another design was the use of a hexapod for ground motion [15]. It has the same concept as the previous design but uses extra actuators to be able to have more control when moving on the ground. The Pens-FlyCrawl [16] is another vehicle that uses a unique 'crawling' mechanism to move on the ground.

All the above mentioned examples have their rotary mechanism (mainly propellers) exposed to the surroundings and there is not much of a 'guard' to protect the moving objects. Hence, a different set of approach is also made where these components are safe inside a cage-type structure. This does not only keeps the propellers safe but also keeps the surroundings from getting in contact with any moving parts. The HyTAQ (Hybrid Terrestrial and Aerial Quadrotor) [17] is amongst the first to have a setup where the whole system is covered by a cage. The cage does not have separate

actuation but rather uses the same motors that are used for flying. Bearings are used so that the cage will have passive motion independent of the frame inside it. Inspired by this concept, another similar design was made by [10] who had a cylindrical cage. Their main focus was on the detection of leaks in industrial HVAC ducts and had a variety of sensors attached to it. It also emphasized on the importance of autonomous navigation as the metal ducts act as a Faraday cage, obstructing RC radio signals. Similar to these passive cages another commercial drone is the Parrot Rolling Spider/minidrone [18]. In their paper, they use PID and LQR control techniques to control the drone and then compare between the two methods. One of the most recent designs is the Drivocopter [19]. It uses independent cages that also work as wheels for ground motion. Unlike the other passive cages, these wheels/cages are actuated from a different motor so that it can be controlled more precisely when on the ground.

1.3.2 Adaptive control for quadrotors

In [20], Yohanes and Izabela present a reference guide for UAV applications in indoor environments. They explain how UAVs can be used in material handling and monitoring tasks in manufacturing environments. Moreover, McCabe et.al explains in [21] how UAVs have potential to facilitate indoor construction tasks that are otherwise mundane and challenging. Naturally, important challenges also have to be faced when working with quadrotors. They are severely underactuated since they have six degrees of freedom (three translational and three rotational) and only four independent inputs which correspond to the motor speeds. This makes the resulting dynamics highly nonlinear [22] as the rotational and translational motions are coupled. Moreover, complicated aerodynamic effects also must be considered if modelling and simulation are to be done accurately. Hence, controlling a quadrotor is one of the major tasks in the field of aerial vehicles. There are multiple factors that can effect the control of a quadrotor in flight including change wind variation, mass variation or even an error in the sensor readings [23–25].

To overcome such problems, various control methods have been developed to control the quadrotor autonomously. Some of the early ones are the PID control [26–28] and the LQR control [29, 30]. These methods are simple to implement but have lim-

ited performance as they are vulnerable to external disturbances and noise [22]. Other non-linear methods such as the sliding mode control (SMC) [31, 32] and backstepping control [33] are also used as they can be very efficient. However, these methods require to have an accurate model and correct system parameters. To overcome this, researchers usually combine different methods or use adaptive techniques [34–37] to achieve robust control. Since a quadrotor is extremely susceptible to external disturbances, [38] uses Linear Active Disturbance Rejection Control (LADRC) and compares the result of inducing a disturbance with the Model Reference Adaptive Control (MRAC) method. [37] uses MRAC but does not account for noisy input and gain scheduling. DMRAC can also be used for systems that do not have an exact known model. This could include quadrotors with various extensions or modifications that are added to achieve a different task. [10] shows one of these modifications that changes the system model.

1.3.3 HVAC ducts

The main objective of HVAC ducts is to distribute airflow from the HVAC equipment to a commercial building, industrial area or even an entire house. It involves the air in the building to be sucked into the ducts, transmitted to the main air conditioner/heater and cooled/heated accordingly. In an evaluation survey for energy saving effects based on extended energy consumption and energy efficiency [39], it is noted that air-conditioning in several countries is not an option but a necessity to make living conditions possible. Due to this, the largest portion of electricity consumption in these countries is through air conditioning. Moreover, any leakage or blockages significantly effects the efficiency of the system and hence increases operational costs. Up to 33% [40, 41] of the energy used for air ventilation is lost if there are any leakages or blockage in the ducts. This makes testing of leakage inside ducts an important issue. In a study, 24 houses were monitored before and after repairing any duct leakages that were found [42]. As a result, the energy consumption of the air conditioners reduced by 18%.

Developing this idea further, maintaining leaks would not be possible if they are not detected in the first place. Maintaining a ventilation system can be expensive and, in some cases, it can be dangerous to even localize the leak itself [43]. This is due to the

fact that some regions in the ventilation system can be impossible to physically access by a human. Inspections should be scheduled after certain amounts of time which can result in the job to be dangerous as well as cumbersome. Due to this, some companies have emerged specializing only in duct inspection and maintenance. Some of them also use technologies such as tethered rovers with a camera equipped to access otherwise inaccessible locations.

1.3.4 Current solutions for ventilation maintenance

Variety of options are present for detection of leaks in HVAC ducts. It ranges from as simple as physically inspecting the ducts from the inside after wearing special equipment and clothing [1] to sending a tethered rover with a camera attached to it [2, 3, 44]. As studies [45, 46] show, in-pipe inspection methods are more accurate when compared to outside-of-the-pipe method. This can also be said for HVAC ducts. The option for physically inspecting the duct from the inside is, for obvious reasons, very dangerous. It does not only include the danger of suffocation but also risks the person hurting themselves. Moreover, it depends on constraints such as the size of the ducts throughout the installed length. Access to the ducts is another issue that needs to be considered. Vertical ascension inside the ducts is threatening due to the slippery surfaces. In commercial or industrial areas, ductwork spans to many meters (not including multiple floors). Since inspecting ducts take time, as small leaks are not directly visible to the naked eye, the air inside can become unbearable. If the ventilation system is on during the inspection, then hot/cold temperature for prolonged periods of time can also be another factor. All of this only applies to ducts that have the capability and supports to hold the weight of a person and not collapse.



Figure 1. Physical inspection inside HVAC ducts [1]

Another method of detecting leaks, that is currently being used, is by sending a small rover with an RGB camera attached to it [2, 3, 44]. This has its own advantages and disadvantages. One main advantage is that due to its small dimensions, it can access ductwork that are physically inaccessible. On the other hand, all the available rovers are tethered. This means that navigation inside the ducts must be done manually through a remote attached to the tether. Detection of leaks also needs to be done manually which will be restricted within the field of view of the camera. Any leak or crack not on the camera's view will not be detected. The range of the tether is also limited (20 to 30 meters) which means that the range for the rover to navigate is also limited. Furthermore, the risk for tether entanglement is also present if the tether is very long. Since an RGB camera is used, external lighting, such as LED lights, must also be present on the rover. Another main disadvantage is that all of them are 'wheeled' rovers which means vertical ascension is not possible. Floors will be done independently if there is access open for the rovers to go through. Moreover, they cannot detect any leaks or cracks on the duct that connects two floors since the rover will not be navigating through it.



Figure 2. Left-to-Right: Robosoft Inspection Robot [2], Danduct Inspection Robot [3]

It is difficult to detect thermal leaks with natural human senses. Therefore for detecting thermal leaks, various methods can be used [47]. These include acoustic methods and optical systems. The optic cable sensor is another method that can be used for gas leak detection [48]. However, this can be expensive to implement and needs another gas to be continuously supplied inside the system. Another passive method to detect a gas leak is to use a thermal camera. As long as there is a gas leakage in the system, there will be a temperature change in that area [48]. Nawrat et. all [49] use a low cost thermal imaging camera and present a contactless method for gas leak detection.

Chapter 2.Hardware Review and Selection

This chapter will cover the current hardware available and some basics of their workings. It will go over the workings of the IMU, INS, LiDAR and the stereo camera. Moreover, it will compare between some of these items and the selection of the items will be justified. This is then followed by the selection of the electronics that goes on the drone. These include the motors, electronic speed controllers (ESCs), power distribution board (PDB), propellers, the controller and the companion computer.

2.1. Inertial Measurement Unit (IMU)

2.1.1IMU basics and error

A combination of inertial sensors mounted in a unit that gives the measurement of the rotation and the acceleration in all three axes is known as the Inertial Measurement Unit (IMU) [5]. It consists of two orthogonal sensor triads. One contains three accelerometers and the other has three gyroscopes. The gyroscopes help measure the angular motion on three orthogonal directions, whereas linear motion on the same direction is measured by the accelerometers. Since these sensor triads share the same axes, they have the same origin. Moreover, the whole unit is attached to a fixed body and these axes will be referred to as the body axes or the body frame. The components of the IMU with the common axes are shown in the Figure3:

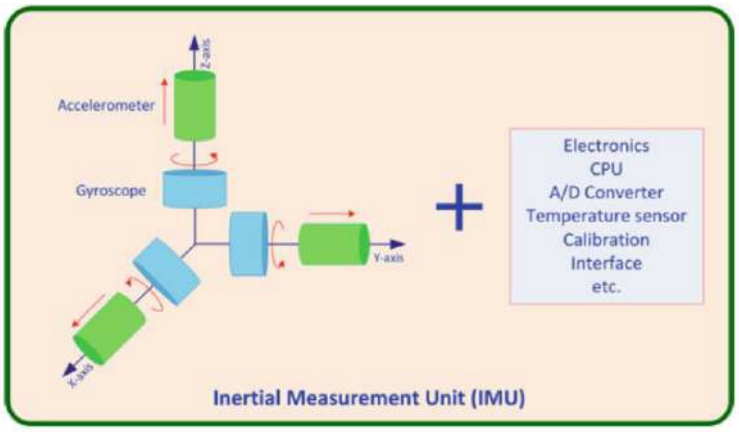


Figure 3. A typical IMU [4]

IMU is usually used with a navigation computer that processes the information from the gyroscopes and accelerometers. This computer will also have initial conditions of the body the IMU is attached to as well as the gravity model. From these inputs and the output from the accelerometer, velocity and position is calculated using integration. On the other hand, the gyroscopes are used to measure the attitude of the body. A block diagram on the overview of this mechanism is shown in Figure4.

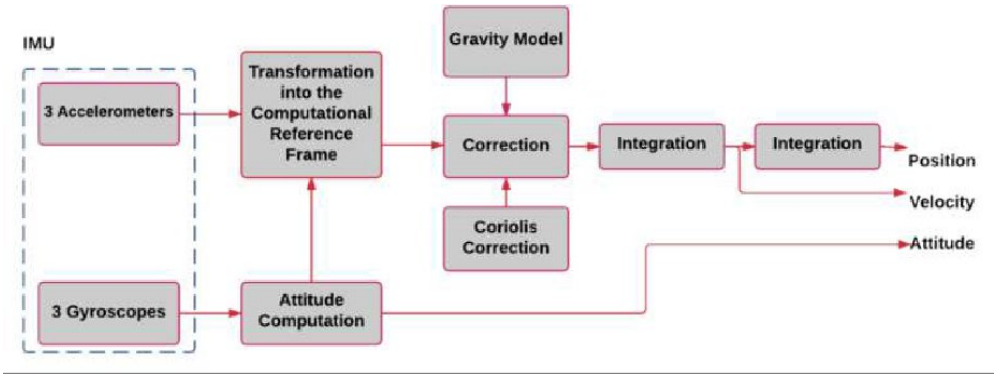


Figure 4. INS block diagram [5]

2.1.2Physical implementation of an INS

As mentioned in [5], there are two main physical implementation of an INS system that can be implemented:

- Stable platform system (gimbaled system)
- Strapdown system

The main difference between the two systems is that one of them (gimbal system) is fixed to its own inertial frame while the other is fixed to the frame of the body it is attached to. The accelerometers and gyroscopes of the gimbal system are fixed in on a stationary plate whose pose and direction is relative to the inertial or earth navigation frame. This is installed inside an outer gimbal that is attached to the vehicle's body. This makes the inner gimbal being isolated from the motion of the vehicle. For it to be stable, torque actuators are used with servo feedback [5]. Due to the complexity of the system and its design, they usually have high accuracy and costs when compared to the strapdown system. On the other hand, due to the physical mechanism it can only

measure small deviations. Another drawback to the system is that it can suffer from gimbal lock where it loses one degree of freedom [50]. This can be fixed by adding an extra gimbal to the setup but then adds to the cost of the system. Strapdown Inertial Navigation System, as the name suggests, is strapped directly to the body of the vehicle and moves along with it. Rate gyros and accelerometers continuously measure the angular motion and the accelerations. These systems are much cheaper when compared to the gimbal system. However, more complex analysis needs to be done to the output of the sensors to get the position, velocity and the attitude of the vehicle. These analyses become more complicated if the pose is required in the earth and navigational frames. Figure 5 shows an overview of both the systems:

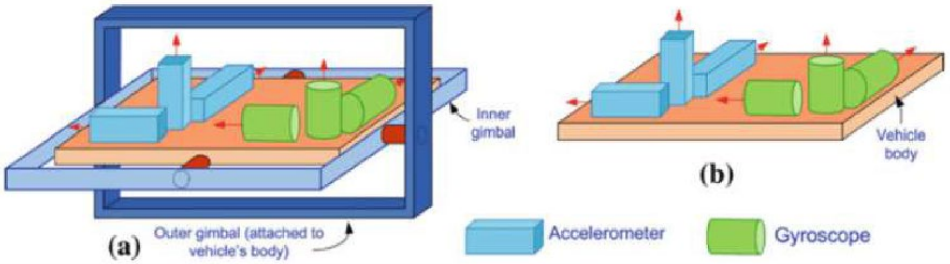


Figure 5. (a) Gimbaled system, (b) Strapdown system [5]

2.2. Light Detection and Ranging (LiDAR) Sensor

Light Detection and Ranging (LiDAR) is a laser-based sensor that help measure distances to a target. A pulse of laser light is emitted, and the reflected pulse is measured from the sensor. Using the time it takes for this process to happen and the speed of light, distance from the target can be calculated. As noted by [51], these sensors are commonly used for making high resolution maps in applications like geography, geology, forestry and laser guidance. An application with an airborne LiDAR is shown in Figure 6. Since it uses ultraviolet, visible or near infrared light for its emitted pulse, it can detect a wide range of materials including clouds, rain and non-metallic objects. Since the laser beam is very narrow, the resolutions of the physical features that it detects is very high [6].

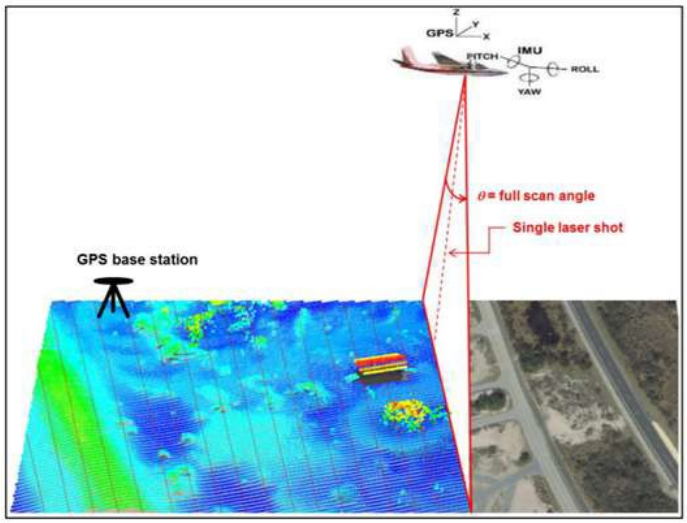


Figure 6. One of the laser scan patterns commonly used for an airborne LiDAR [6]

When this concept of sending and receiving light is done on a plane, the sensor is known as 2D laser scanner. As shown in Figure 7, these laser scanners are mounted on a rotating body and hence a scanning affect is achieved. Moreover, another degree of freedom can be added by letting the sensor scan up-down as well to yield a 3D scanning device [52]. [52] also compares the laser scanner with the sonar. It mentions how most of the time is wasted in a sonar device when waiting for the pulse to return. On the other hand, when using a laser device, the problem arises in how fast we can rotate the device as the pulse speed is extremely high. The angular resolution and accuracy are also far better when compared to sonar sensors. Moreover, data received from the laser sensor can directly be interpreted as the distance to the obstacle in a certain direction; making post-processing less demanding on the equipment it is used with. The main drawback for the laser sensor is the initial cost. Its applications are very specific, and many other sensors can be used in its place which makes the LiDAR expensive.

2.3. Stereo Camera

Stereo cameras use stereo image sensing technology to help calculate the 'depth' of an image. This technology makes use of two cameras and information from both is used to make a three-dimensional image [53]. This is visualized in Figure 8. This setup allows to simulate human binocular vision and therefore helps to perceive depth. The



Figure 7. The RP LiDAR A1 with motor system to rotate 360° [7]

distance between the cameras is fixed and is known as the intra-ocular distance. To create a proper depth image, three elements must be determined [54]. These are known as the foreground, background and the zero parallax. The image that appears closest to the viewer will be known as the foreground; whereas, the image that will appear on the farthest will be the background. The default zone between them is the zero parallax. The closer the object is to the cameras, the more misaligned they will be to the cameras and hence they will have more divergence. With the same principle, the farther they are, they will be more aligned and hence divergence will be less.

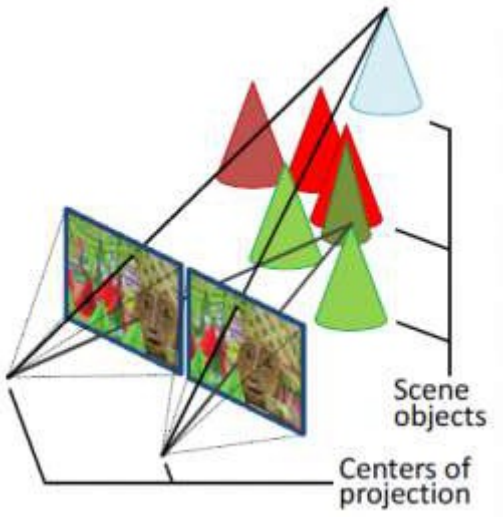


Figure 8. The projection rays for both the cameras [8]

Stereo depth perception can be classified into two types:

- Passive stereo

•Active stereo

Passive stereo cameras do not have their own lighting source and hence depend on external or the environment's light. This suggests that they perform better in areas that are well-lighted and textured. On the other hand, active stereos employ their own light as shown in Figure 9. This can be in the form of laser or any structured light. This will also help the cameras in matching the points for their image and help give a better result. Active stereos are mainly used where there is less light available. Developing this further, it performs the same way as the passive stereo does if sunlight is present or if the effect of the light source is negligible due to extra light. Moreover, in long-ranges it will work similar to the passive stereo as the light source will not be effective. The IR projector or the light source will also add to the cost of the overall camera.

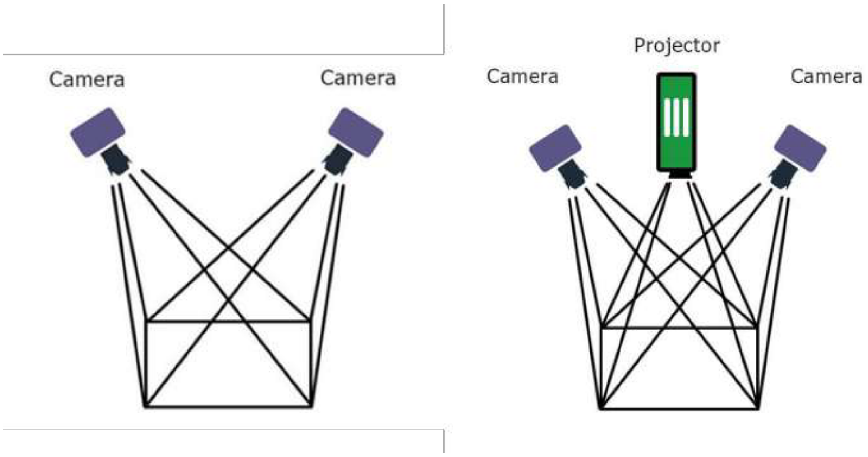


Figure 9. Active and passive stereo cameras [8]

2.4. Comparison Between LiDAR and Stereo

For measuring the distance around the quadrotor, either the LiDAR or the stereo camera can be used. An A1 model RP LiDAR and an Intel D435 stereo camera is available in the Mechatronics lab and hence these models will be compared with each other. Table 1 compares some parameters between the two including the cost for both of them:

Table 1. Comparing A1 RP LiDAR with the Intel D435 stereo camera

	A1 Rplidar [7]	Intel D435 Camera [53]
Field-of-View (Horizontal)	0 – 360 Degrees	91 Degrees
Field-of-View (Vertical)	0 Degrees	65 Degrees
Dimensions (mm)	98.5x70x60	90x25x25
Operating Range (Min-Max)	.15m – 8m	.11m – 10m
Angular Resolution	$\leq 1 Degree$	$\approx \leq 0.1 Degrees$
Weight	170 g	72 g
Measuring frequency	2000 Sa/s	30 FPS
Accuracy (distance from object)	$\leq 0.5\%$	$\leq 1\%$
Intensity of light required	Low	High
Post-processing time	Low	High
Price	$ 99.00	$ 179.00

As discussed in Section 2.2, the LiDAR can be made to rotate on a motor such that it can get data points in a plane rather than in one direction only. This concept is used by the A1 RP LiDAR and hence its FOV is 360° in the horizontal plane. On the other hand, the stereo camera will be fixed on a particular direction and its FOV is limited to 91° (Figure 10). The LiDAR can be made to get readings on the vertical axis as well by adding an extra motor that will allow it to rotate on the movement plane. However, in our application it is not really required to get data points in the vertical direction. In both cases, changes in the yaw for the quadrotor can be made from the readings acquired from the sensors. In the case of LiDAR it is easier since it measures distances on the whole plane. The distances on the immediate right and left can be used to know the attitude of the quadrotor and fix if necessary. For the stereo camera, the readings from the edges can be used to have an estimate of the attitude. These readings need to be similar if the vehicle direction needs to be the same as that of the duct direction.

Measurement frequency of the LiDAR can be increased by increasing the speed of the motor that rotates the receiver and laser transmitter. Having more measurements per time means that the values from the IMU readings will be corrected more frequently and hence the overall values will be close to the true values. Moreover, the processing speed required on the LiDAR is low compared the stereo camera and hence will be less extensive on the microcontroller. If placed in the middle of the drone, the minimum

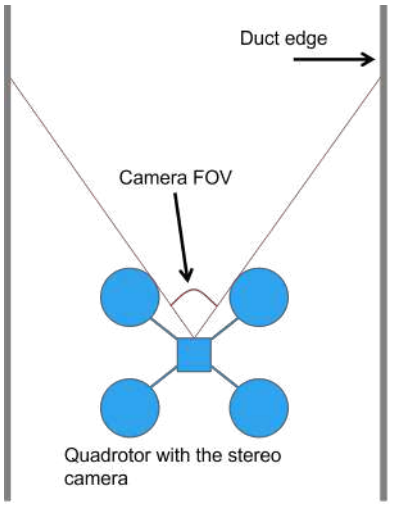


Figure 10. Top view of the stereo camera inside the duct

distance that both the LiDAR and the stereo camera can possibly measure should be lower than the width of the rudimentary ducts made. This will allow the same ducts to be used for testing purposes. Since the width of the ducts are around 740 mm, the minimum distance (when placed in the middle to the edge of the duct) is 370 mm. This lies within the operating range of both the sensors and hence the rudimentary ducts can still be used for testing. Both the sensors were also placed inside a duct with limited lighting. This was done to visualize the readings that they can produce before installing them on a drone.

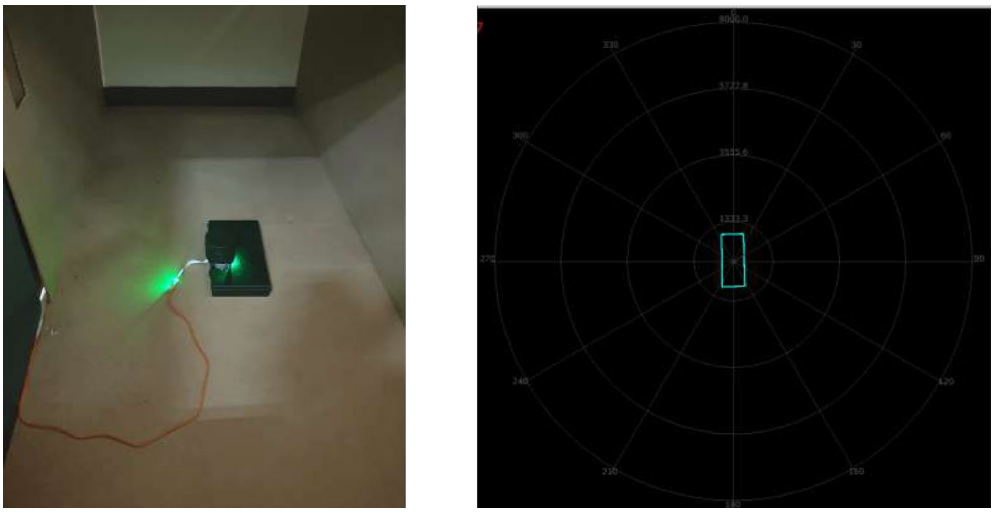


Figure 11. Placing the RPLidar inside a rectangular duct with limited lighting

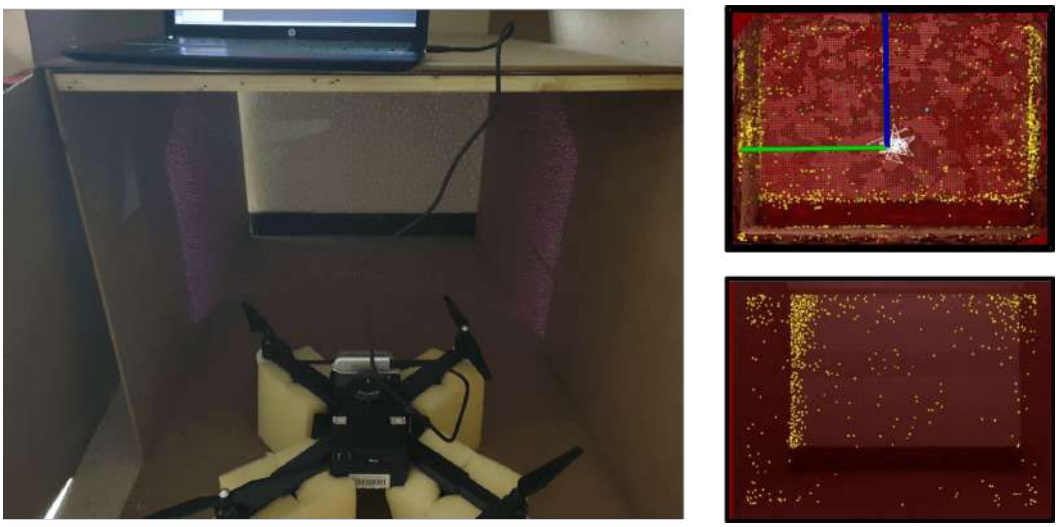


Figure 12. Placing the Intel D435 inside a rectangular duct with limited lighting

Figure 11 shows the RPlidar inside the duct and Figure 12 shows the Intel D435 Stereo camera inside the duct. On the left sides of the figures is that actual hardware placement. On the right side are the readings obtained when these sensors and ran. Notice that the LiDAR produces a 2D plane of the readings that are consistent with the actual dimensions. On the other hand, the camera produces limited results. The yellow dots are the ones that the camera recognizes and the red area is unknown. This is mostly because of the limited lighting inside the duct as well as the lack of features present. Because of these results and the discussion mentioned above, the LiDAR was selected to be used on the drone as opposed to the stereo camera.

2.5. Thermal Camera

A thermal camera is a type of camera that detects infrared radiation and allows objects with high temperatures to be distinguishable from a distance. It is helpful in high-end applications such as fire-fighting services that allows them to detect areas of heat through smoke or any other visibility barrier [55]. They include components such as an optic system and IR detectors which are also the main reason why these cameras are expensive. Due to this, thermal imaging cameras come in various sizes. The number of pixels generated for the image is reduced in order to compensate for the cost of some of them. For drone applications, a major requirement for any on-board equipment them is to be light-weight. They also have to be compact in size as to not obstruct

any moving equipment on the drone. The AMG8833 [56], shown in Figure 13, is a compact temperature sensor that is compatible with the Raspberry Pi. It generates 64 pixels and has a temperature range of 0°C to 80°C. It weighs only 23 grams and has I2C communication to generate real time thermal image.



Figure 13. AMG8833 - Thermal Camera

2.6. Optitrack

An Optitrack is a system for motion capture which involves the process of recording the movement of objects or people. Also known as motion tracking, motion capture has various biomedical applications [57]. Along with animation and game development, this kind of technology is also used for gait analysis and facial tracking.

The system uses a set of cameras that are projected at a specified field (known as capture volume). These cameras are tuned to capture specific markers. The markers are coated with retro-reflective material such that they can be differentiated from the objects around them. These kind of markers are known as passive markers and they do not require any extra wires or electronic equipment. The cameras are first calibrated with known markers in order to adjust lens distortion of each camera.

2.7. Selection of Electronic Components for the Drone

This section will go over the basic electronic components that are needed for the drone to operate. This will include the microcontrollers as well as the motors and propellers used. It will also highlight how these parts are connected with the battery so that each component is able to draw enough current for itself.

2.7.1 Flight controller and companion computer

Pixhawk was selected as the flight controller with Raspberry Pi being a companion computer. The pixhawk on its own is an independent open-hardware project that provides standards for numerous hardware designs for winged, multi-rotor and also wheeled vehicles [58]. Another advantage is that it is able to communicate with the Raspberry Pi as its companion computer such that the computer is the hub for various sensors which then commands the pixhawk on what the vehicle has to do. Additional hardware such as the GPS, compass and the telemetry receiver can also be directly connected to the pixhawk with its UART serial ports. The pixhawk also comes with an inbuilt IMU (MPU6000 as main accel and gyro) that will be used for the drone's navigation.

2.7.2 Motors, ESCs and propellers

The initial design of the drone included a set of 4 motors. A frame size of 250 mm (diagonal motor center-to-center) was selected so that it can fit easily inside an industrial duct. This sets the maximum propeller size to be around 5 inches in diameter. However, to increase the lift of the drone so that it is able to lift a LiDAR sensor along with a rolling cage attached to it, the number of motors were increase to 8 that would be attached co-axially with the previous system. The coaxial configuration helps to keep the drone of the same dimensions without having it increase in size.

The selection of the motors were done based on the propellers. Since the propellers are not big in diameter, their weight is also less which allows us to use lower rated motors. Motors of size 1800kV (HobbyWing) were selected. Four of them facing upwards with the propellers mirrored diagonally and four of them facing downwards

with the propellers mirroring the motors above them. This configuration is known as an octa-quadcopter. Based on the size of the motors, ESCs could be selected for 30 amps (maximum current the motor can take).

The pitch of the propellers were also changed and experimented with over time. Initially, high pitch propellers (5 inch pitch with four wings) were used which in theory helped increase the lift of the overall drone. However, for the case of an octa-quadcopter, the higher the pitch, the more unstable the drone becomes. This then allowed to reduce the pitch to 3.7 inches and the wings were increased to four. A simulation was conducted on E-Calc [9] to check the overall composition of the parts selected. These results are shown in Figure 14.

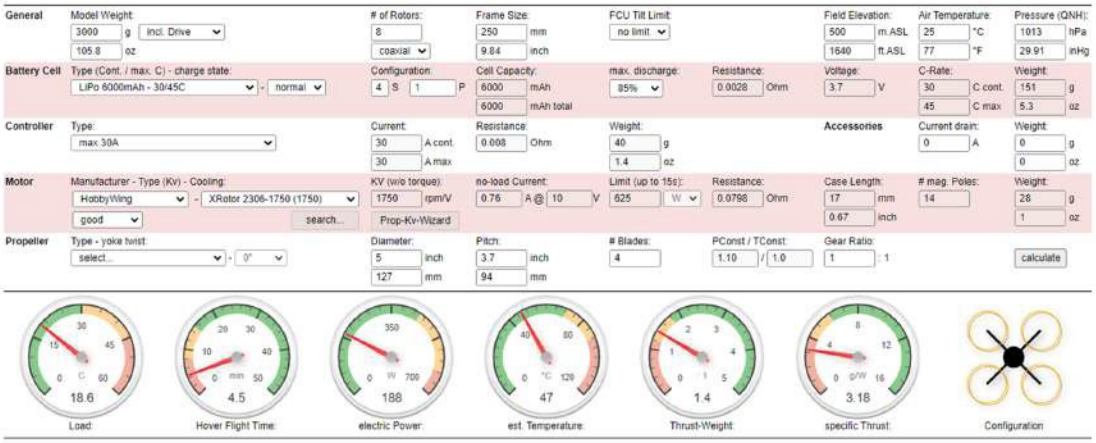


Figure 14. Simulation results from ecalc [9] for drone electronics

2.7.3 Power distribution board (PDB), battery and electronics assembly

A big power distribution board with 8 channels was used to support all the 8 motors. This also had an extra 5V DC supply for the flight controller. This is shown in Figure 15.

The power to the raspberry-pi was not given through the PDB. Instead an extra voltage step-down module was used that decreased the voltage to 5V needed by the raspberry pi. This was done mainly for the LiDAR to work correctly. The power for the LiDAR comes from the raspberry-pi. If the power for the raspberry-pi was given through the PDB, then the current supplied to the LiDAR would not be enough for it

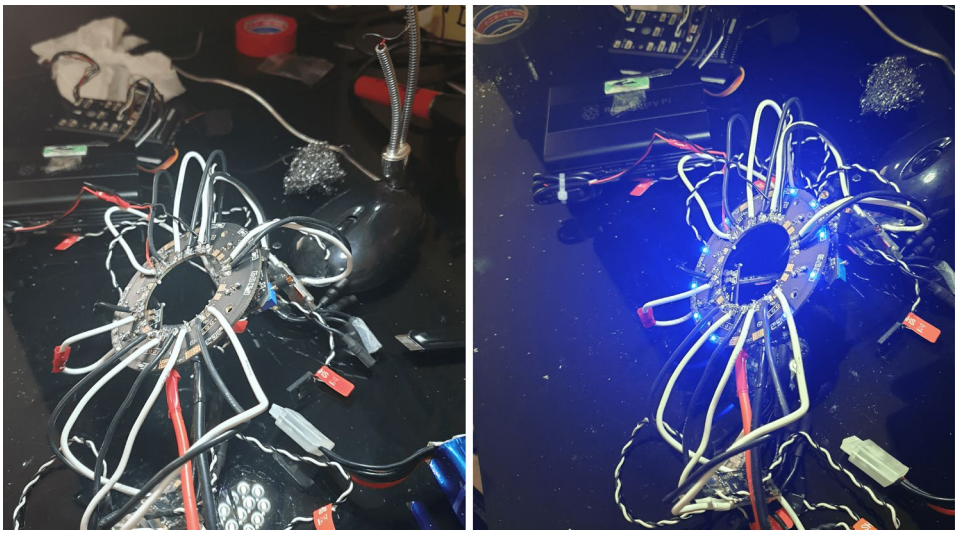


Figure 15. Power distribution Board with power off and on

to work properly. This is why an extra channel for the voltage was provided to it. A summary of the electronics connection is shown in Figure 16.

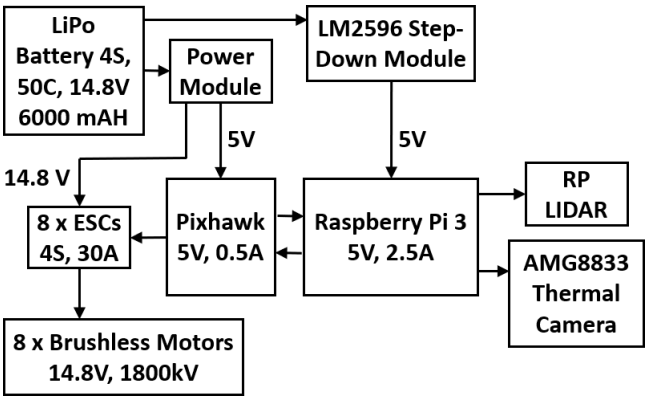


Figure 16. Electronics connection configuration for the drone

A 4 cell 14.8V LiPo battery was used with a current rating of 6000mAH, 50C. This allows enough burst current for all the motors and the extra electronic components to work properly.

Chapter 3.Design and 3D Printing

This chapter involves the design and 3D printing of the vehicle frame as well as the cage around it. It explains the necessary components required for the installation of the electronics as well as the idea to attach a rolling cage on top of the main frame. It also covers the 3D printing process and how everything was assembled together for the final system.

3.1. Drone Frame Design

The design of the frame should be such that it is capable to hold all the electronics and the weight of the overall vehicle is balanced. Moreover, it should be versatile enough such that the electronics can easily be assembled and disassembled if needed. Since the LiDAR constituted to be a major part of the weight of the drone (along with the battery), both of these components were placed at the bottom of the frame. The flight controller (Pixhawk 4) is placed in the dead center of the vehicle (as required by the controller itself). In addition to these, an extra layer is added so that the companion computer (Raspberry Pi) and other electronics can fit inside. The overall frame of the vehicle is shown in Figure 17.

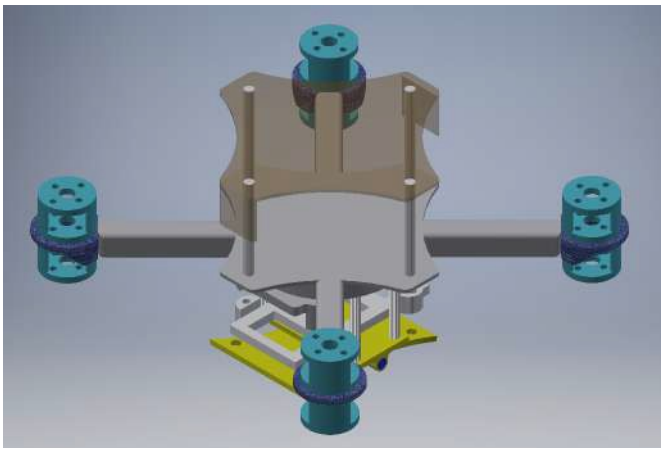


Figure 17. Frame of the vehicle without electronics

The blue parts at the edges of the vehicle in Figure 17 (called motor extenders) allows additional motors to be attached at the bottom of the rover. This allows the drone

to be an octaquadrotor rather than just a quadrotor. More details about the connections of the eight motors and their directions is covered in Section 2.7.2. Since the LiDAR is placed at the bottom of the drone, it has to be such that it matches the orientation of the IMU at all times. To achieve this, the LiDAR is connected rigidly with the body of the vehicle. A pin is used to easily assemble and disassemble in case the battery needs to be replaced. This is shown in Figure 18 which is the side view of the frame.

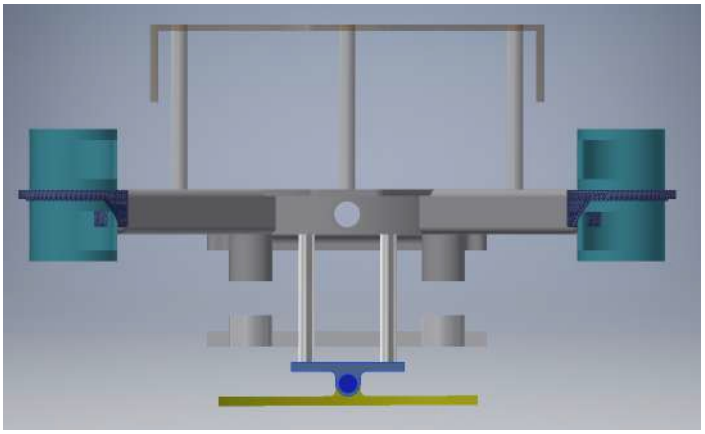


Figure 18. Side view of the vehicle

3.2. Cage Design

The design of the cage should be such that the drone 'rolls' when it is on the ground. It should also provide a protective cover for the moving parts of the drone. In addition to that, it should not hinder the free movement of the drone and be able to rotate independently on its own. The design was inspired by concept of a gyroscope that has rings around it to move freely. The main frame was mounted on a shaft with bearings on both ends which had the cage attached to them. This allowed the motion of the cage to be independent of the inner frame. One iteration of the cage is shown in Figure 19.

As seen from Figure 19, the static stress analysis showed that large deformations would occur when a load equal to the predicted weight of the drone (3 kgs) was added to this iteration of the cage. To overcome this, the thickness of the cage was increased to help cushion any impact the drone goes through. The final cage design is shown in Figure 20. This also shows all the electronics, including the LiDAR and the motors,

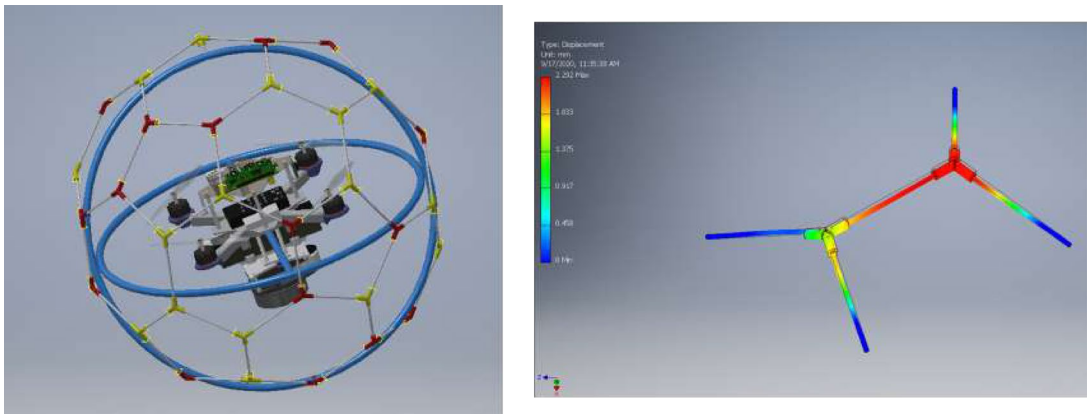


Figure 19. First iteration for cage design

fitted on the design with the 'gyroscopic' ring around it. The shaft was selected to be made of solid aluminum so that its deflection is minimum and it does not have a lot of weight on its own.

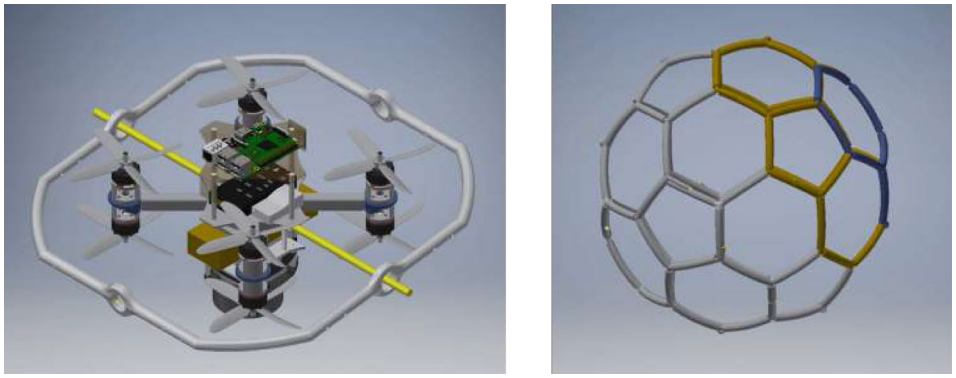


Figure 20. Final iteration for cage design

3.3.3D Printing and Overall Assembly

After the design process, the parts were then 3D printed. The material chosen for the parts was ABS. This material provided enough rigidity for the cage to not loose its shape on an impact and enough elasticity to withstand an impact. It was printed with nozzle temperature of 240°C with bed temperature of 100°C. The infill was kept at 20% to keep the cage light enough for the drone to move. Due to the nature of the design, there were not a lot of supports for the print and hence the printing was done with supports on. To save time, multiple parts were printed at one go as shown in Figure 21.

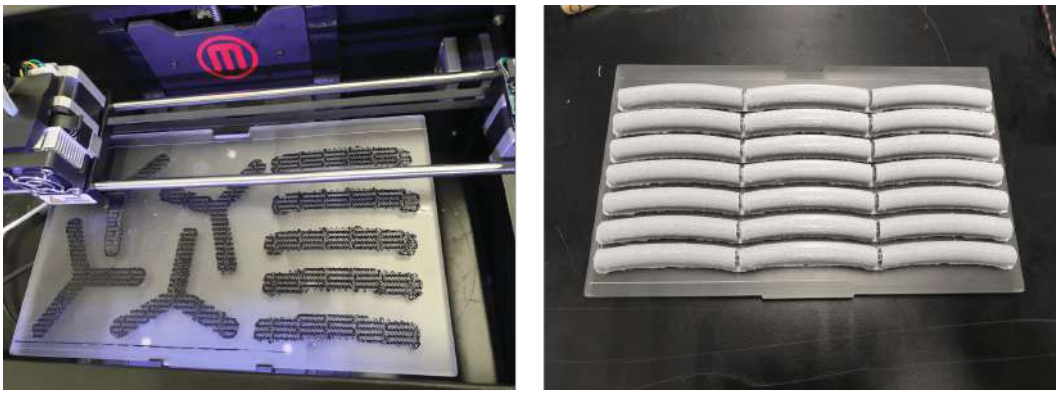


Figure 21. 3D printing cage components

The final setup is shown in Figure 22 which includes all the printed and the electronic components. Drone specifications are shown in Table 2.

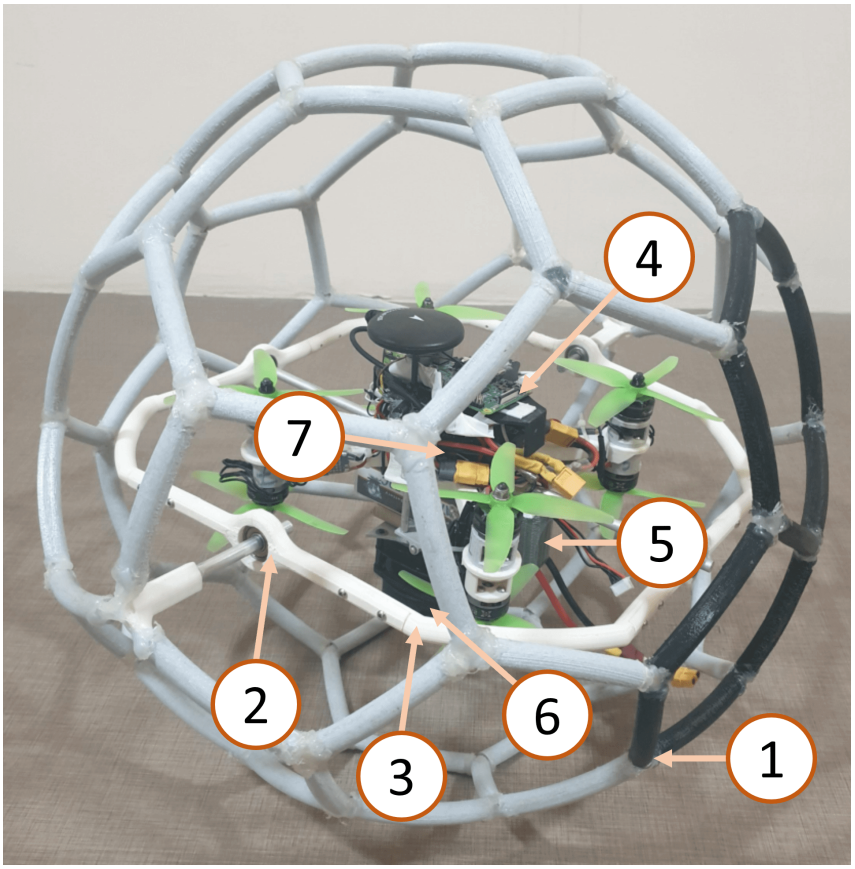


Figure 22. Main drone components: 1 - 3D printed assembled cage, 2 - Bearing for the inner ring to rotate on, 3 - Inner ring, 4 - Raspberry Pi for high level control, 5 - Battery, 6 - LiDAR, 7 - Pixhawk with IMU for low level control

Table 2. Drone specifications

Cage diameter	50cm
Drone dimensions - hub-to-hub	25 cm
Flight time	4 min
Rolling time	15 min
Rolling speed	2.75 m/s
Weight	3 kgs
Propeller length	5 in
ESC size per motor	30 A
Battery size	6000mAH, 50C, 4S

The weight of the vehicle was measured to be around 3 kgs. A ground speed test was conducted to measure the speed of the vehicle (Figure 24). Based on the distance covered (with placed landmarks) in particular time (measured through video frames as shown in Figure 24), the speed was calculated to be 2.75 m/s. The 'x', 'y' and 'z' coordinate frames are shown in Figure 23.

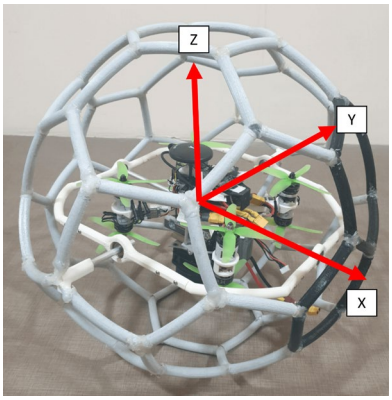


Figure 23. Coordinate frames for the drone



Figure 24. Frames from a video captured for the drone's ground speed test

Chapter 4.Simulation Design and Setup

In this chapter, Direct Model Reference Adaptive Control is used coupled with PID controllers to control the position of a quadrotor. Gaussian white noise with standard deviation of 5 is used to model noisy sensors that measure the body rates of the system. Moreover, four techniques to combat the drifting of adaptive gains are used. These include dead-zone modification, sigma modification, epsilon modification and a projection based modification. The chapter also investigates the changes in the gains when some parameters in the adaptive laws are changed.

Section 4.1.1 presents the quadrotor model used in the simulations. This is derived in detail in Appendix A. Section 4.1.2 shows the adaptive controller with the equations of all the gain modifications. Section 4.2 shows the simulation setup. Section 4.3 shows the simulation results of the gains for all the gain modifications. Section 4.3.2 shows the effect of changing the ε parameter in the dead-zone modification on the adaptive gains. Section 4.3.3 shows the effect of changing the σ parameter in the sigma modification on the adaptive gains. Section 4.3.4 shows the effect of changing θ_{max} parameter in the projection based modification on the adaptive gains. The error graph comparing the actual and the reference model is also plotted for all cases.

4.1. Problem Statement

4.1.1 Quadrotor model

The quadrotor model used in the simulation is as follows [28]:

$$\ddot{X} = (\sin\Psi\sin\Phi + \cos\Psi\sin\Theta\cos\Phi)\frac{U_1}{m} \tag{1}$$

$$\ddot{Y} = (_\cos\Psi\sin\Phi + \sin\Psi\sin\Theta\cos\Phi)\frac{U_1}{m} \tag{2}$$

$$\ddot{Z} = _g + (\cos\Theta\cos\Phi)\frac{U_1}{m} \tag{3}$$

$$\dot{p} = \frac{I_{YY} - I_{ZZ}}{I_{XX}} qr - \frac{U_2}{I_{XX}} \tag{4}$$

$$\dot{q} = \frac{I_{ZZ} - I_{XX}}{I_{YY}} pr - \frac{U_3}{I_{YY}} \tag{5}$$

$$\dot{r} = \frac{I_{XX} - I_{YY}}{I_{ZZ}} pq - \frac{U_4}{I_{ZZ}} \tag{6}$$

where X, Y, Z are the position of the center of mass WRT inertial frame, φ, θ, ψ are the Euler angles, p, q, r are the body rates, m is the Quadrotor's mass, I_{XX}, I_{YY}, I_{ZZ} are the moments of inertia and $U1, U2, U3, U4$ are the throttle, roll, pitch and yaw forces and moments respectively. The mass m is set to be $3kgs$ and the moments of inertia are achieved based on the quadrotor model in Autocad:

$$I_{XX} = I_{YY} = 0.2880, I_{ZZ} = 0.5280 \tag{7}$$

The dynamics of the actuator are defined as:

$$U_1 = b\left(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2\right) \tag{8}$$

$$U_2 = bl\left(-\Omega_2^2 + \Omega_4^2\right) \tag{9}$$

$$U_3 = bl\left(-\Omega_1^2 + \Omega_3^2\right) \tag{10}$$

$$U_4 = d\left(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2\right) \tag{11}$$

where b is the thrust factor, d is the drag factor, l is the radius of the quadrotor and Ω_i is the propeller's speed of motor i.

4.1.2 Controller design

A Direct Model Reference Controller (DMRAC) along with a Proportional-Integral-Derivative (PID) controller is used to control the quadrotor. The PID controller is defined as follows:

$$U_i \quad K_P e(t) + K_i \int_0^t e(\tau)\, d\tau \quad + K_d \frac{d}{dt} e(t) \tag{12}$$

where e is the error defined as the difference between the predefined target and the actual value. A total of 6 PID controllers are used where the PIDs for x, y, pitch and

roll are cascaded together as shown in Figure 25. Figure 26 shows the control loop for the yaw control.

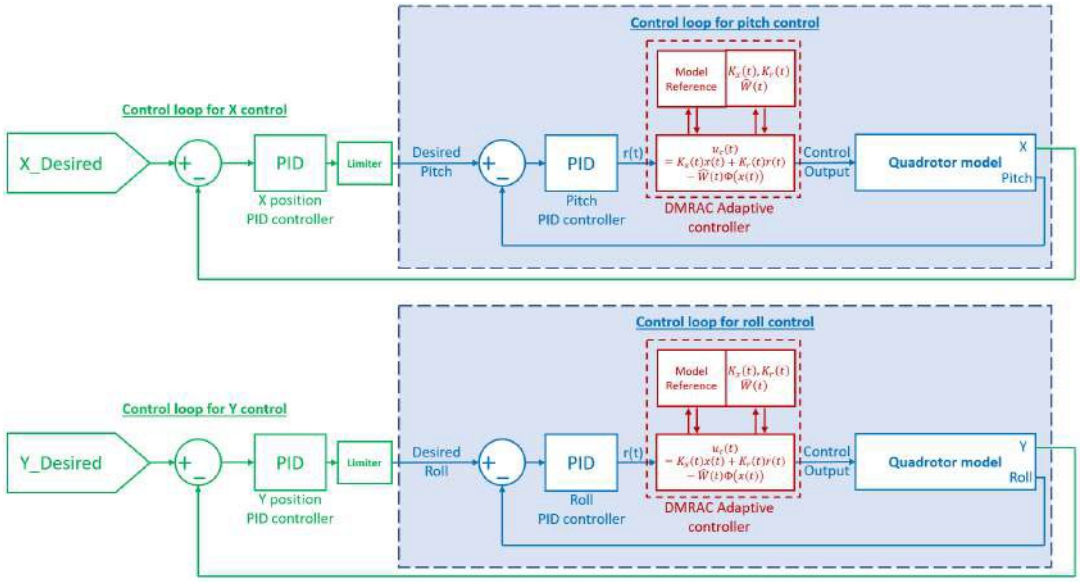


Figure 25. Control loop for x, y, pitch and roll motion

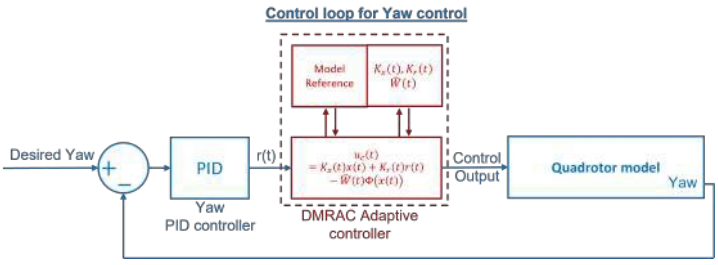


Figure 26. Control loop for yaw motion

The Direct MRAC scalar case is used for controlling further the body rates of the quadrotor. An additional error signal with a standard deviation is added to the body rate equations in order to simulate an actual quadrotor sensor.

$$\dot{p} = \frac{Iyy - Izz}{Ixx} qr - \frac{U2}{Ixx} + std_{dev} * rand(1) \tag{13}$$

$$\dot{q} = \frac{IzZ - Ixx}{Iyy} pr - \frac{U3}{Iyy} + std_{dev} * rand(1) \tag{14}$$

$$\dot{r} = \frac{Ixx - Iyy}{I_{ZZ}} pq - \frac{U4}{I_{ZZ}} + std_{dev} * rand(1) \tag{15}$$

As the position of the quadrotor closely depends on the body rates of the system (as seen in equations 1 - 3), the main objective is to control these three variables as precisely as possible. Therefore, scalar case DMRAC is used to control these three variables. The model system used for the DMRAC is shown in equation 16. A PID controller in the inner loop is added before the signal goes to the DMRAC controller. This is done to help set the steady state behaviour of the system; whereas, the model reference help set the transient response behaviour for the system.

$$x_m(t) = A_m x_m(t-1) + b_m \; r(t) + k_u \Delta u(t) \quad , x_m(0) = 0 \tag{16}$$

$$\Delta u(t) = u(t) - u_c(t) \tag{17}$$

where:

$$u(t) = u_{max} sat(\frac{u_c(t)}{u_{max}}) \tag{18}$$

$$u_c(t) = K_x(t)x(t) + K_r(t)r(t) - \hat{W}(t)\Phi(x(t)) \tag{19}$$

A_m and b_m are scalar constants. For all three cases, the reference model is kept the same such that:

$$A_m = -1.5, b_m = 0.5 \tag{20}$$

$u_c(t)$ is the adapted input to the system and u_{max} is the maximum input allowed to the motors. $\Phi(x(t))$ is the matched nonlinear function that is different for the three body rates. These are shown in equations 21 -23.

$$\Phi(x(t))_p = qr \tag{21}$$

$$\Phi(x(t))_q = pr \tag{22}$$

$$\Phi(x(t))_r = pq \tag{23}$$

k_x, k_r, k_u and $\hat{W}$ are adaptive gains for the DMRAC. These gains are calculated by their own differential equations. Different modifications are added to these equations in order to avoid these gains to drift. Four modifications are used as shown below:

4.1.2.1. Adaptive laws with no modifications

With no modifications, the differential equations can be written as:

$$\dot{k_x}(t) = -\gamma_x x(t)e(t)\,\mathrm{sign}(b),\ k_x(0) = 0 \tag{24}$$

$$\dot{k_r}(t) = -\gamma_r r(t)e(t)\,\mathrm{sign}(b),\ k_r(0) = 0 \tag{25}$$

$$\dot{k_u}(t) = \gamma_u \Delta u(t)e(t)b_m,\ k_u(0) = 0 \tag{26}$$

$$\dot{\hat{W}}(t) = \gamma_w \Phi(x(t))e(t)\,\mathrm{sign}(b),\ \hat{W}(0) = 0 \tag{27}$$

4.1.2.2. Adaptive laws with dead-zone modification

The dead-zone modification basically makes the velocity of the gains to be zero if absolute of the error falls below a certain predefined value 'ε'.

$$\dot{k_x}(t) = \begin{cases} -\gamma_x x(t)e(t)\,\mathrm{sign}(b), & \text{if } |e| > \varepsilon \\ 0, & \text{if } |e| \leq \varepsilon \end{cases} \tag{28}$$

$$\dot{k_r}(t) = \begin{cases} -\gamma_r r(t)e(t)\,\mathrm{sign}(b), & \text{if } |e| > \varepsilon \\ 0, & \text{if } |e| \leq \varepsilon \end{cases} \tag{29}$$

$$\dot{\hat{W}}(t) = \begin{cases} \gamma_w \Phi(x(t))e(t)\,\mathrm{sign}(b), & \text{if } |e| > \varepsilon \\ 0, & \text{if } |e| \leq \varepsilon \end{cases} \tag{30}$$

$$\dot{k_u}(t) = \begin{cases} \gamma_u \Delta u(t)e(t)b_m, & \text{if } |e| > \varepsilon \\ 0, & \text{if } |e| \leq \varepsilon \end{cases} \tag{31}$$

4.1.2.3. Adaptive laws with sigma modification

This is similar to adding a damping term to the adaptive laws such that the velocity of the gains is reduced. The constant 'σ' is predefined.

$$\dot{k_x}(t) = -\gamma_x x(t)e(t)\,\mathrm{sign}(b) + \sigma_x k_x(t),\ k_x(0) = 0 \tag{32}$$

$$\dot{k_r}(t) = -\gamma_r r(t)e(t)\,\mathrm{sign}(b) + \sigma_r k_r(t),\ k_r(0) = 0 \tag{33}$$

$$\dot{k}_u(t) = \gamma_u \Delta u(t) e(t) b_m + \sigma_u k_u(t),\ k_u(0) = 0 \tag{34}$$

$$\dot{\hat{W}}(t) = \gamma_w \Phi(x(t)) e(t)\,\mathrm{sign}(b) + \sigma_w \hat{W}(t),\ \hat{W}(0) = 0 \tag{35}$$

4.1.2.4. Adaptive laws with epsilon modification

This increases the damping term by adding the absolute of the error value to it. The damping will be higher if the error is higher.

$$\dot{k}_x(t) = -\gamma_x x(t) e(t)\,\mathrm{sign}(b) + |e|\sigma_x k_x(t),\ k_x(0) = 0 \tag{36}$$

$$\dot{k}_r(t) = -\gamma_r r(t) e(t)\,\mathrm{sign}(b) + |e|\sigma_r k_r(t),\ k_r(0) = 0 \tag{37}$$

$$\dot{k}_u(t) = \gamma_u \Delta u(t) e(t) b_m + |e|\sigma_u k_u(t),\ k_u(0) = 0 \tag{38}$$

$$\dot{\hat{W}}(t) = \gamma_w \Phi(x(t)) e(t)\,\mathrm{sign}(b) + |e|\sigma_w \hat{W}(t),\ \hat{W}(0) = 0 \tag{39}$$

4.1.2.5. Adaptive laws with projection based modification

This kind of modification uses projection based approach. If the value of the gain is higher than a predefined norm-bound, then the value is reduced based on the difference between the allowed and the actual value. The laws are defined as follows:

$$\dot{k}_x(t) = \gamma_x Proj(k_x, -x(t) e(t)\,\mathrm{sign}(b)),\ k_x(0) = 0 \tag{40}$$

$$\dot{k}_r(t) = \gamma_r Proj(k_r, -r(t) e(t)\,\mathrm{sign}(b)),\ k_r(0) = 0 \tag{41}$$

$$\dot{k}_u(t) = \gamma_u Proj(k_u, \Delta u(t) e(t) b_m),\ k_u(0) = 0 \tag{42}$$

$$\dot{\hat{W}}(t) = \gamma_w Proj(\hat{W}, \Phi(x(t)) e(t)\,\mathrm{sign}(b)),\ \hat{W}(0) = 0 \tag{43}$$

where the projection operator is defined as:

$$Proj(\theta, v) = \begin{cases} v, \text{ if } \psi(\theta) < 0 \\ v, \text{ if } \psi(\theta) \geq 0 \,\&\, < \nabla\psi, v > \geq 0 \\ v - \psi(\theta) < \frac{\nabla\psi}{\|\nabla\psi\|}, v > \frac{\nabla\psi}{\|\nabla\psi\|}, \text{ else} \end{cases} \tag{44}$$

$$\psi(\theta) = \frac{\theta^T\theta - \theta_{max}^2}{\varepsilon_\theta \theta_{max}} \tag{45}$$

θ_{max} is the norm-bound imposed on θ and ε_θ is the convergence tolerance.

4.2. Simulation Setup

The simulation is done on MATLAB using its scripts and functions. The predefined environment is set as shown in Figure 27. Six points are set as the desired path for the quadrotor to pass through. Another simulation setup was also made to resemble a duct such that the drone travels inside it. The first part of Figure 28 shows the duct as a whole and the second part removes some aspects of the duct to view the inside points more clearly. It also monitors and displays the distance readings of the drone while the program is running.

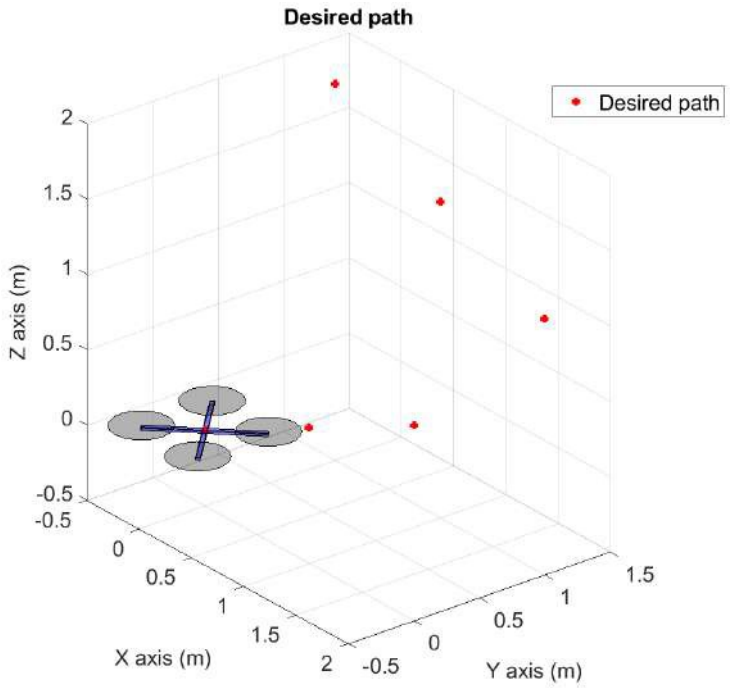


Figure 27. Desired path for the quadrotor

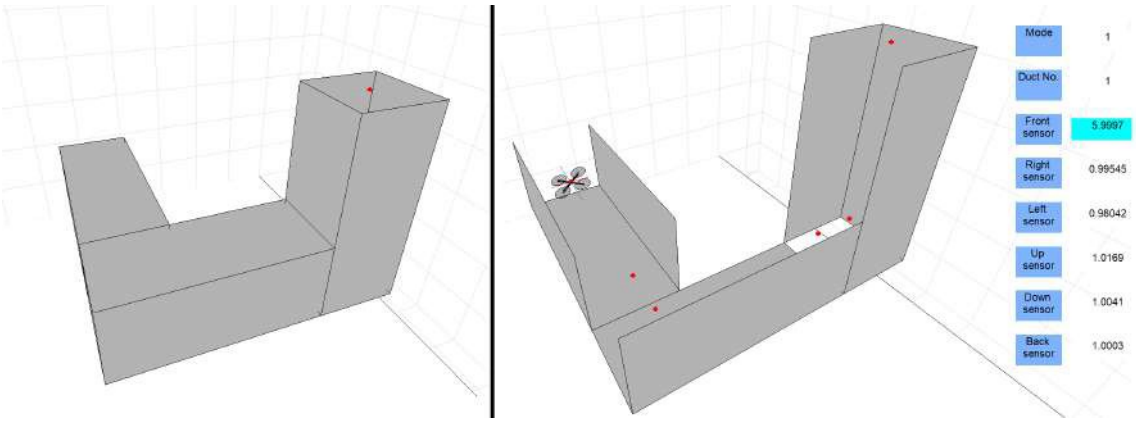


Figure 28. Simulation setup for drone inside a duct

The standard deviation for the Gaussian white noise is set to 5 and the gains for the adaptive laws are set as shown in equation 46

$$\gamma_u = \gamma_w = 2\gamma_r = 4\gamma_x = 80 \tag{46}$$

Similar to the reference model, the gains for all three body rates are kept the same as the equations have the same structure. Euler Integration is used for solving all the differential equations. The simulation can be seen in real time as its position and body rates are translated/rotated throughout the experiment. Three main graphs are acquired from the simulation. The first one is the performance of the system. This shows the 'x', 'y' and 'z' positions of the quadrotor with respect to time. This graph looks identical for all cases as the target is kept the same (Figure 29). The second important plot is the adaptive gains values over time. This is to see if the gains drift over time and what changes happen to them when the modifications are applied. The last plot is the error equation (equation 47) which is the difference between the measured body rate and the reference model. Results and discussion for the second and the third graphs are shown in Section 4.3

$$error = x(t) - x_m(t) \tag{47}$$

where $x(t)$ is the body rate (output of equations 13 - 15) and $x_m(t)$ is the reference model (equation 16).

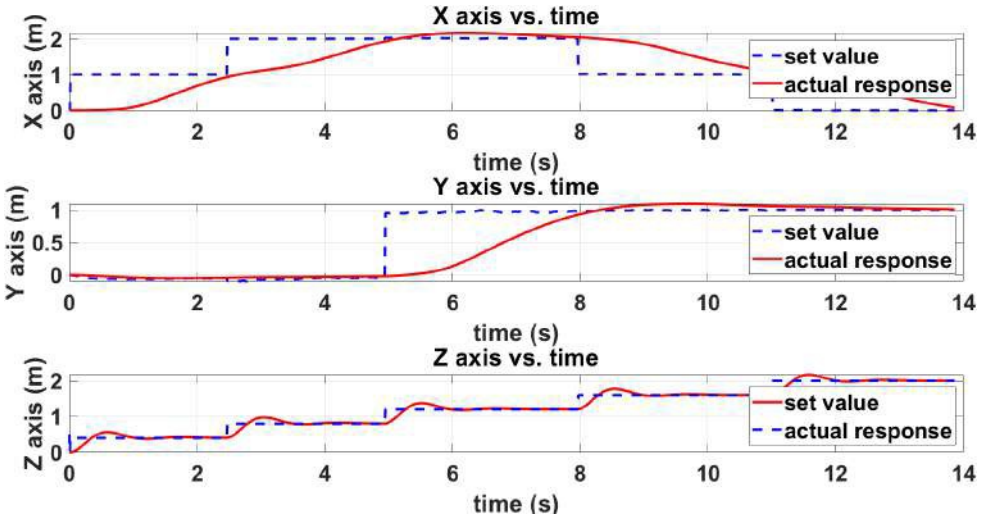


Figure 29. System response

The values of the PID gains are set as follows:

$$X_{Kp} = Y_{Kp} = 0.0837; X_{Ki} = Y_{Ki} = 0.01; X_{Kd} = Y_{Kd} = -0.0175 \tag{48}$$

$$Z_{Kp} = 5.882; Z_{Ki} = 1.01; Z_{Kd} = 6.06 \tag{49}$$

4.3. Simulation Results

This section is divided into four main parts. The first one shows the results obtained for all five cases. The second section shows the difference in the gains for the dead-zone modification if the epsilon in the condition is changed. The third one shows difference in the sigma modification gains when the σ gain is changed. The last section shows the change in the gains for the projection modification when the parameters for the projection function is changed.

4.3.1 Gains and error for all cases

4.3.1.1. No modifications

The system is first run without any modifications to the adaptive gains. Equations in Section 4.1.2.1 are used and the gains and error for each body rate versus time is shown in Figures 30 and 31. The adaptive gains drift after a certain period of time. This

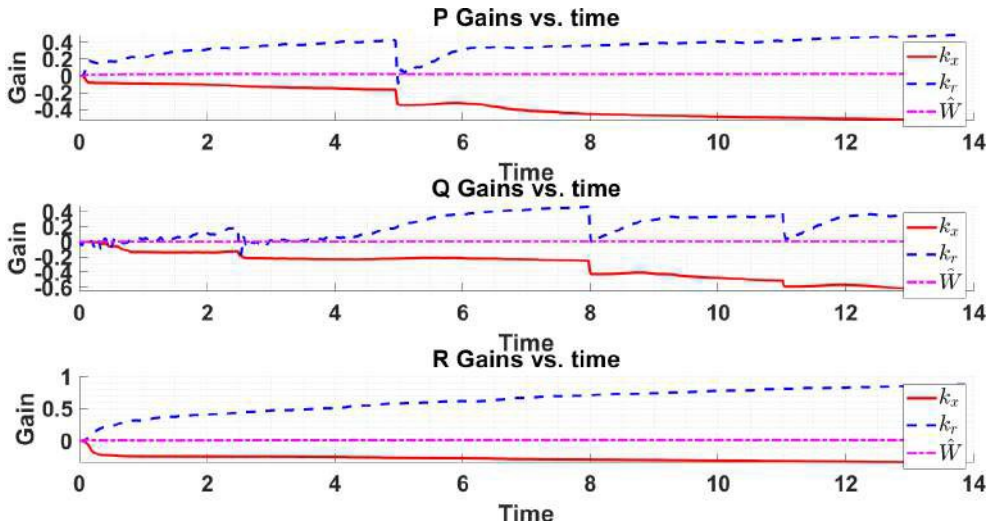


Figure 30. Adaptive Gains versus time with no modifications

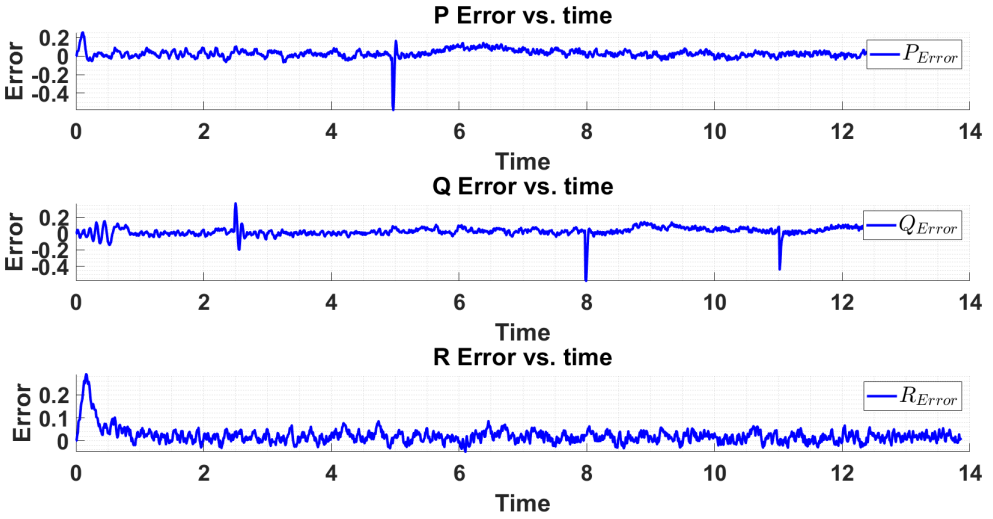


Figure 31. Error versus time with no modifications

is expected as no modifications are added to the laws. The k_x gain increases in the positive direction and the gain k_r increases in the negative direction. The magnitude of the $\hat{W}$ gain is not as high compared to the other two. The error for all three cases fluctuate around zero. The fluctuations are caused because of the random error added to the systems. Since it fluctuates around zero, we can say that our actual system matches the reference model even if a random error signal is added to the system.

4.3.1.2. Dead-zone modification

The equations for the adaptive laws are then changed to the ones shown in Section 4.1.2.2. The ε value is set to 0.2 for all three body rates as that is the average value of

the gains in the previous section. The gains and error versus time are shown in Figures 32 and 33 respectively.

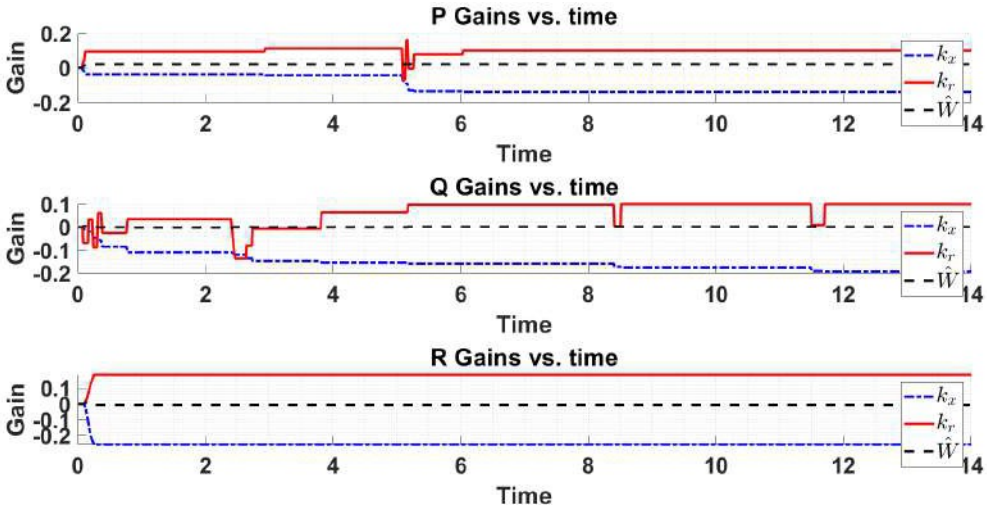


Figure 32. Adaptive Gains versus time with dead-zone modification

With the dead-zone modification, the gains do not drift as much compared to the previous section. However, they do saturate for a specific period of time. The gains change whenever the quadrotor undergoes a change in direction. This could be because of the addition of the noise signal coupled with the sudden change in the desired position. The signs of the gains are the same and they follow similar trends when no modifications are added. As seen in Figure 32, the gains are still increasing over time which shows that the result is still unbounded. Moreover, the error also fluctuates around zero for this case. The spikes in this error is less in the dead-zone modification when compared to no modification. This could be because the velocity of the gains become zero when the error between the model and the reference falls below ε.

4.3.1.3. Sigma modification

The damping term is then added to the adaptive laws as shown in Section 4.1.2.3. The damping coefficient is defined as:

$$\sigma_x = \sigma_r = \sigma_w = 0.2 \tag{50}$$

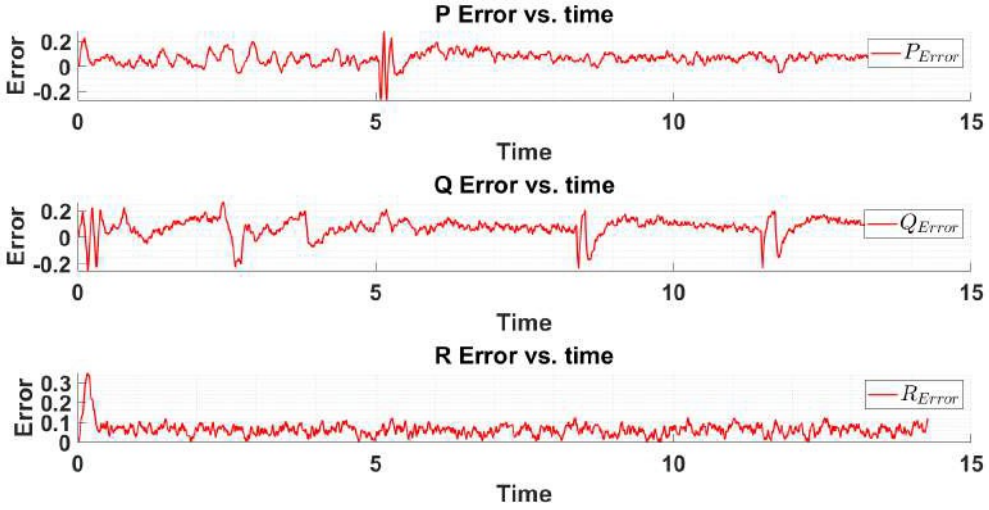


Figure 33. Error versus time with dead-zone modification

This coefficient is kept the same for all three body rates. The gains and error versus time is shown in Figures 34 and 35 respectively.

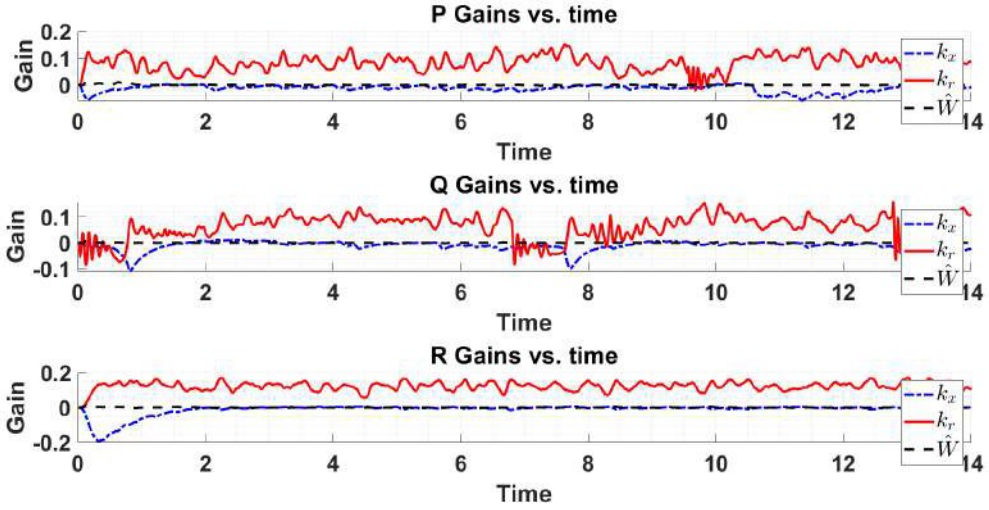


Figure 34. Adaptive Gains versus time with sigma modification

Compared to the previous gain graphs, the gains in the sigma modification fluctuates a lot. However, these fluctuations are bounded and the gains do not drift over time.

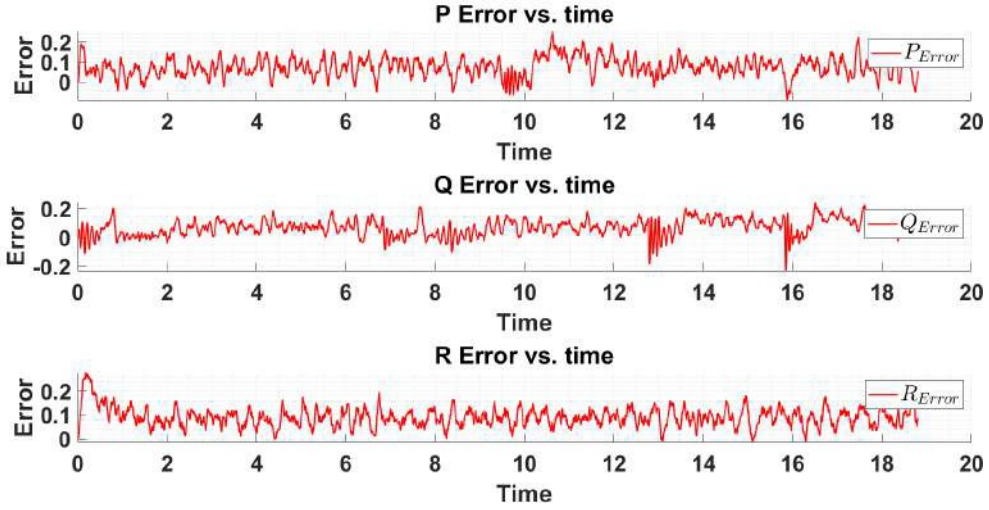


Figure 35. Error versus time with sigma modification

4.3.1.4. Epsilon modification

The absolute of the error is then added to the damping term in the adaptive laws as shown in Section 4.1.2.4. The gains and error versus time are shown in Figures 36 and 37 respectively.

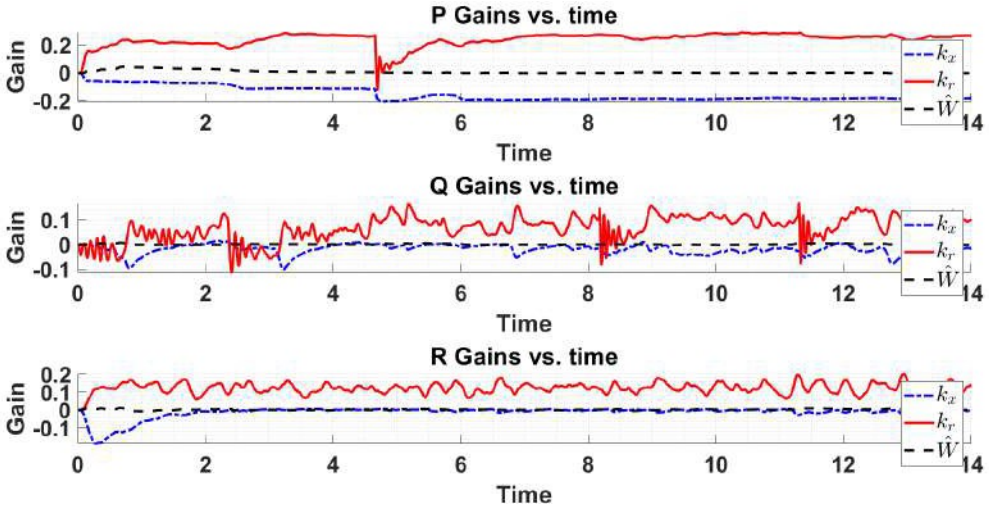


Figure 36. Adaptive Gains versus time with epsilon modification

There is not much difference between the epsilon modification and the sigma modification. This is mainly because the structure for both of them is the same; how-

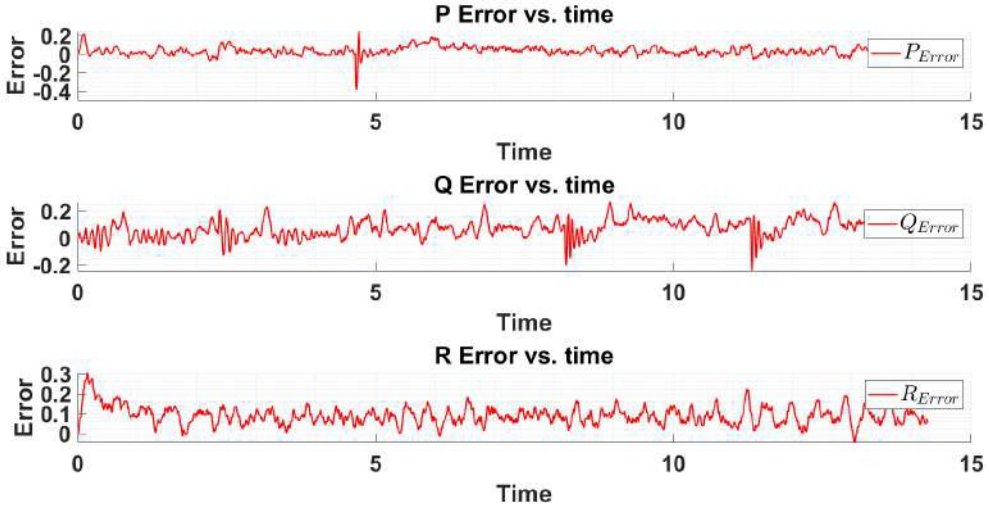


Figure 37. Error versus time with epsilon modification

ever, the damping term is increased based on the error. Similar to the epsilon modification, the gains here are bounded and do not drift over time.

4.3.1.5. Projection based modification

Next, the adaptive laws shown in Section 4.1.2.5 is used. The parameters used inside the projection function are set as shown in equation 51

$$\Theta_{\max} = 0.1, \varepsilon_{\Theta} = 2 \tag{51}$$

The results of the gains and errors are shown in Figures 38 and 39 respectively.

The gains, when applying the projection function, have a strict limit and as seen in Figure 38, they do no pass this limit. The fluctuations in the error is still the same; however, the average magnitude is less compared to the previous cases. This could be because of less fluctuations in the adaptive gains. Since these adaptive gains do not fluctuate as often (compared to the previous cases), the error mainly depends on the noise added to the system.

4.3.2 Changing ε in dead-zone modification

The ε in the dead-zone modification condition is then changed in order to see the difference it makes on the adaptive gains. Two other scenarios were tried; $\varepsilon = 0.1$ (lower

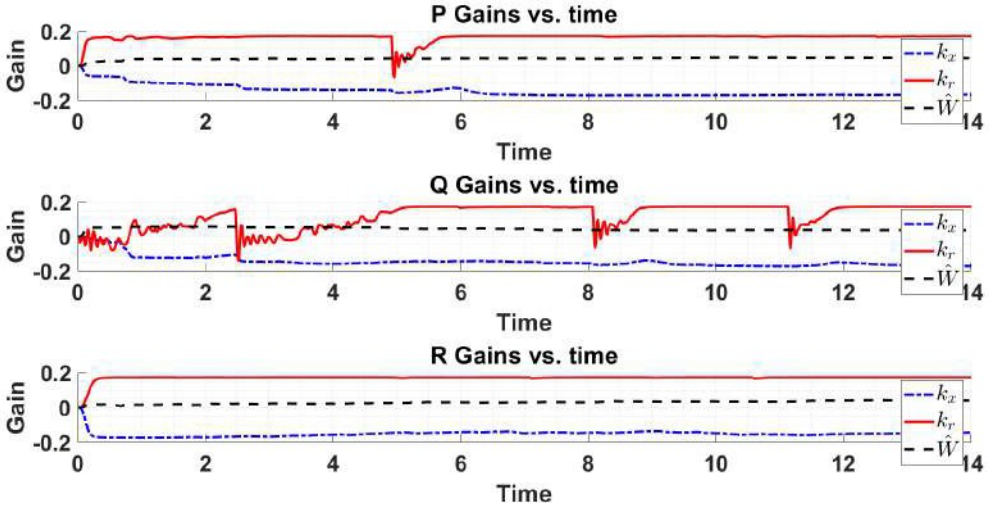


Figure 38. Adaptive Gains versus time with projection based modification

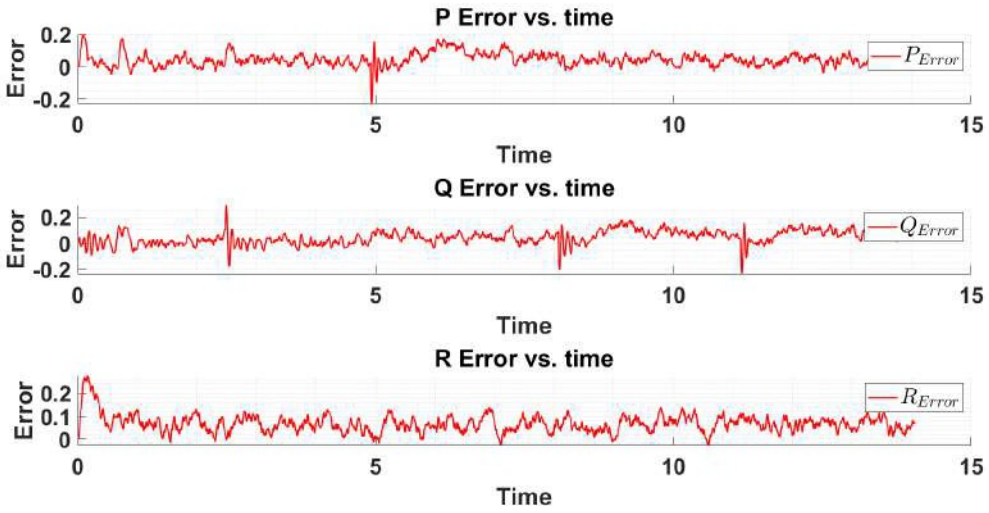


Figure 39. Error versus time with projection based modification

than the previous value) and $\varepsilon = 0.4$ (higher than the previous value). The results were plotted on the same graph as shown in Figure 40.

When the ε is increased, the values of the of the gains saturate at a lower level. This is because the velocity of the gains become zero at a higher value of error and hence saturates earlier. The error graph is also plotted as shown in Figure 41. Since the gains saturate at a lower level when ε is increased, the error is higher with high epsilon.

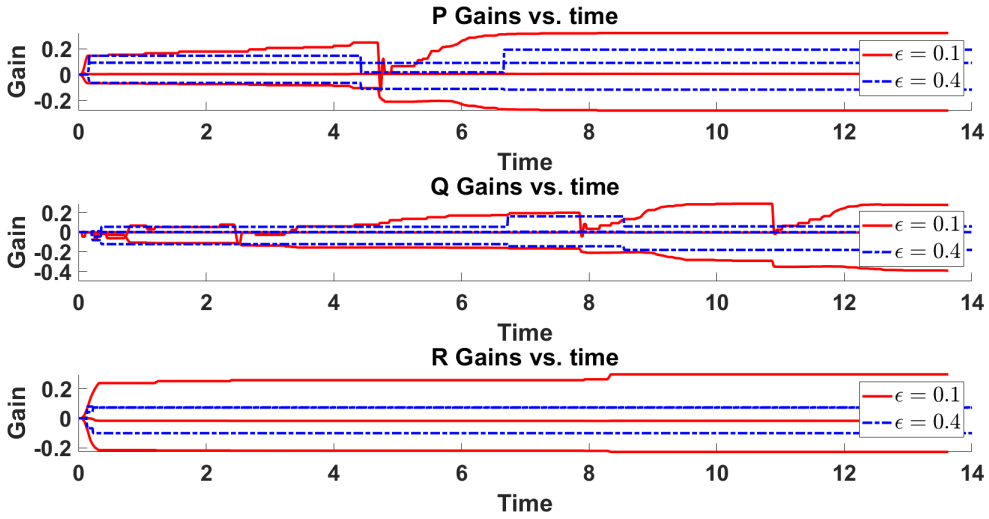


Figure 40. Comparing gains for change in epsilon in the dead-zone modification

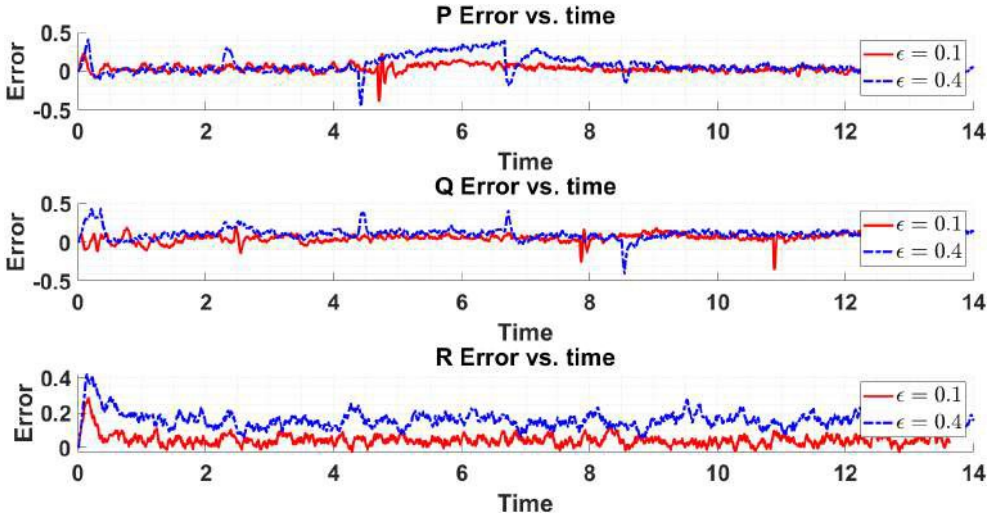


Figure 41. Comparing error for change in epsilon in the dead-zone modification

4.3.3 Changing σ gain in sigma modification

The damping coefficient (σ) is also changed to see its effect on the adaptive gains. Similar to the previous section, the constant is set to 0.1 and 0.4 and the effects in both the gains and the error is seen in Figures 42 and 43 respectively.

When the σ is increased, there is more damping and hence the amplitude of the gains decease. This can clearly be seen in Figure 42. On the other hand, the frequency of oscillations and its amplitude decreased as σ is increased from 0.1 to 0.4 as seen in Figure 43.

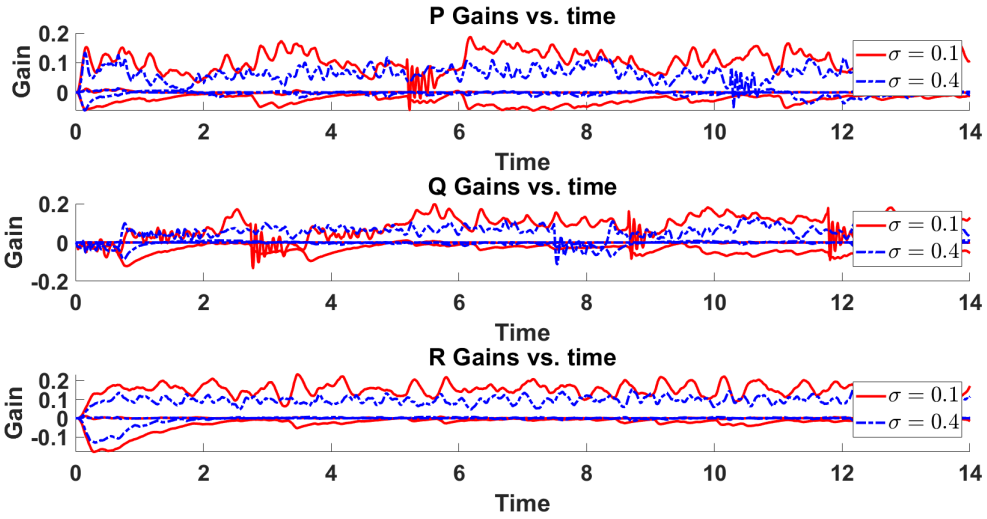


Figure 42. Comparing gains for change in σ in the sigma modification

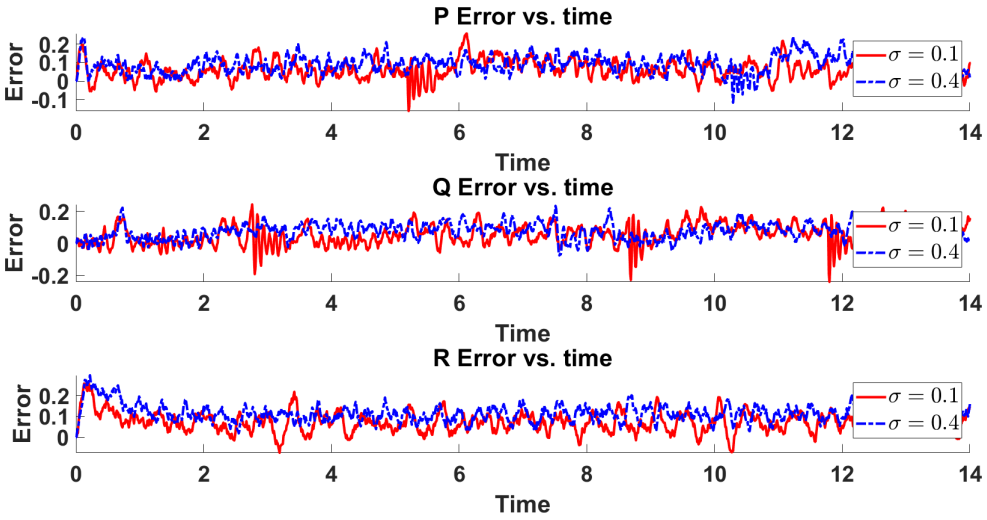


Figure 43. Comparing error for change in σ in the sigma modification

4.3.4 Changing Θ_{max} in the projection modification

Similar to the previous sections, Θ_{max} is both increased (0.2) and decreased (0.05) and the gains and the error is plotted on the same graph to see the difference. These can be seen in Figures 44 and 45 respectively.

Reducing the Θ_{max} reduces the maximum value allowed for the gains. This can be seen in Figure 44 as the $\Theta_{max} = 0.05$ saturates at a lower value compared to $\Theta_{max} = 0.2$. However, this has an effect in the error. The error for a higher Θ_{max} is lower as the gains do not saturate at a lower value as seen in Figure 45. This observation can also be generalized for all cases. If the values of the gains are not allowed to be higher, then the

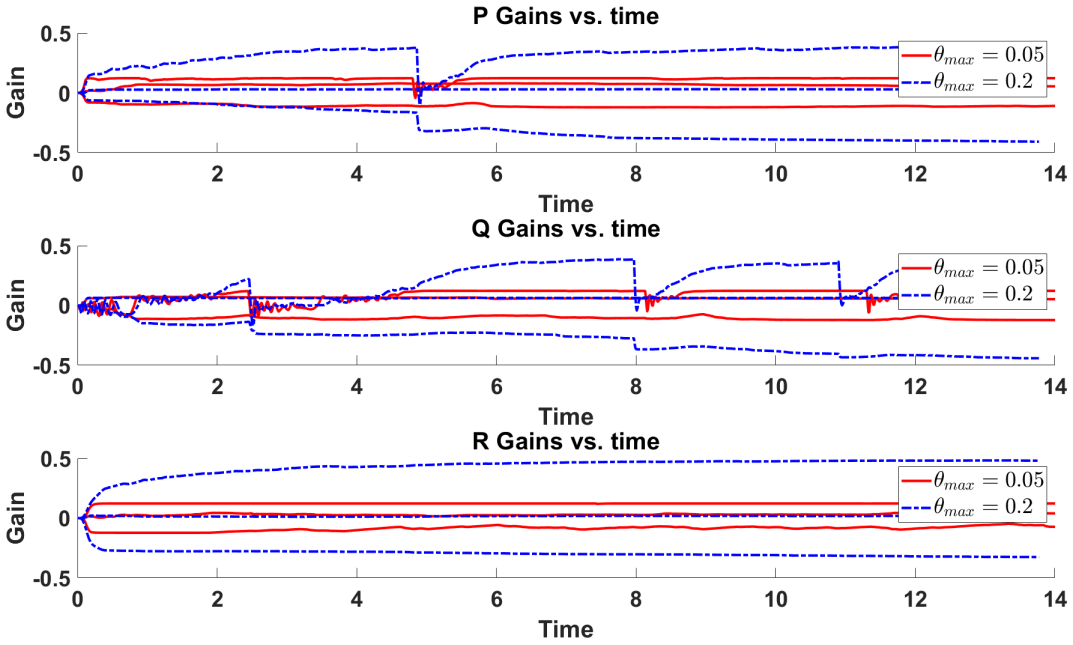


Figure 44. Comparing gains for change in θ_{max} in the projection modification

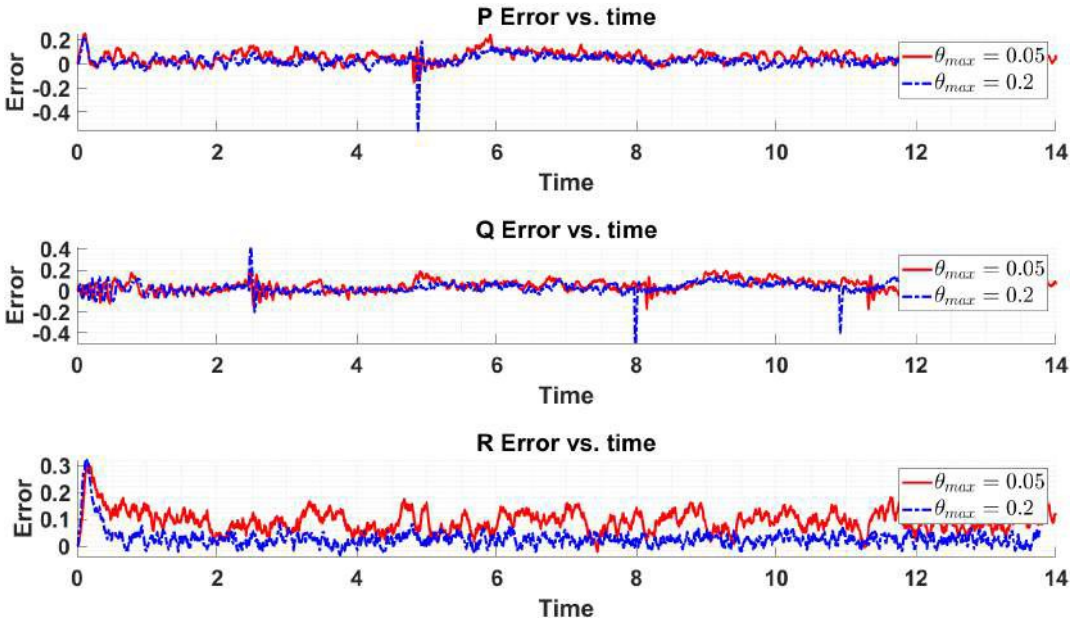


Figure 45. Comparing error for change in θ_{max} in the projection modification

error is seen to increase. However, if the gains are kept to be uncapped, then this can increase the memory requirement for the system and can slow down the control loop.

The same method used in the simulation for the control of the drone was then used in the actual setup described in Chapter 2. Robot Operating System (ROS) was used as the main interface between the micro-controller, flight controller and the LiDAR. A few extra things were added to the system to interface the hardware setup with the software algorithms. One of them was the conversion of the output to PWM signals so that it could be fed to the motors. This was done through a calibration curve established between the angles and the PWM of the motor. Another thing that was added was the LiDAR's projected reading by using the IMU's readings. This was done to get the horizontal distance travelled by the drone. The experiments were only done in one axis for both PID control and adaptive control. The best results of both are then compared with each other.

5.1. Interfacing the Micro-controller, Flight Controller and the LiDAR

ROS [59] was used as the main interface between the electronic components. Each component has its own frame that is then linked to the others. For connecting to the Pixhawk through ROS, the 'mavros' node was similar to the one used by [60]. It is important to note that the stream rate has to be set independently so that the readings of the IMU can be seen by the ROS system. This is done through the following ROS service: "rosservice call /mavros/set stream rate 0 10 1".

For the LiDAR, the 'rplidar' node was used to get the readings. These readings are published as laser scan and the frame was set such that it is rigid with the base frame of the drone but with a fixed translation in the z-axis. This can be visualized as seen in Figure 46.

5.2. Conversion of Output from Control Law to PWM

A calibration curve was established between the angle in radians that the drone can achieve and the PWM command given to the motor. Since the cage can be held firmly and the drone inside can rotate freely, the angle the drone gets can be measured through

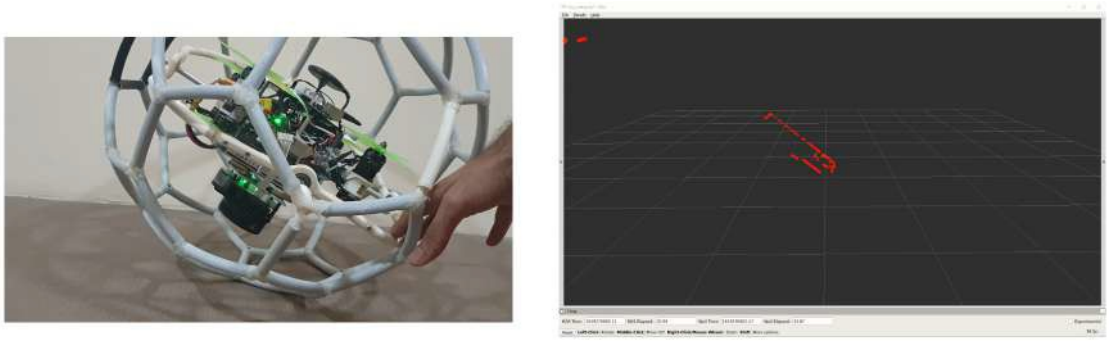


Figure 46. Visualizing LiDAR data

the IMU. So to get the curve, the whole range of PWM that the motor can go (1050-1900) was given and the corresponding angle (read by the IMU) was recorded. This is shown in Figure 47.

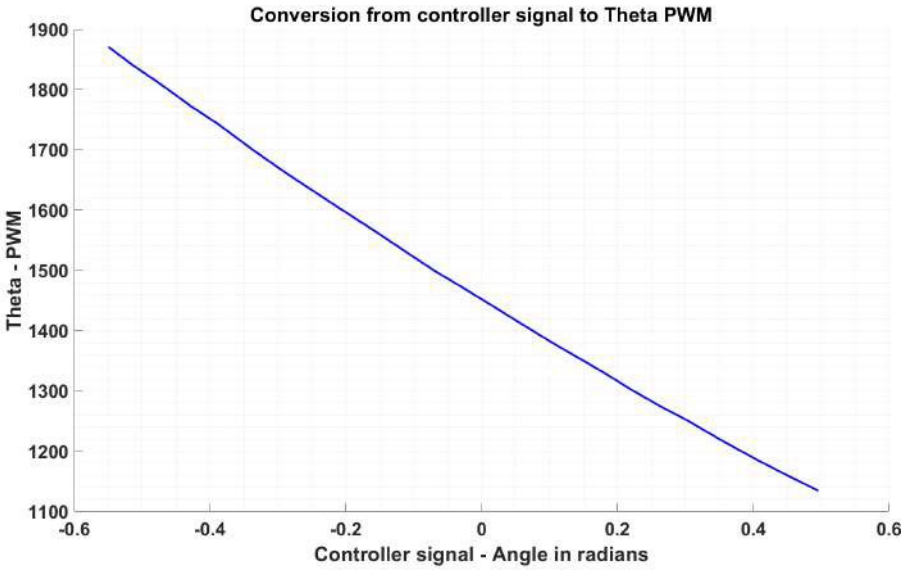


Figure 47. Conversion from controller signal to PWM

To use it in the code the graph was converted to an equation as follows:

$$theta_{PWM} = 85.0045u^2 - 642.18u + 1489.2 \tag{52}$$

where $theta_{PWM}$ is the signal to the motors and 'u' is the output from the control law in equation 19.

5.3. Correcting the LiDAR Data

The raw readings from the LiDAR were first corrected by projecting them in the horizontal plane by using the IMU. Since the control was only done in one plane, the planar distance of the drone was used in the control. Using the IMU, simple cosine rule was used as shown in Figure 48

$$reading_x = (r_{reading})(cos(theta_r)) \tag{53}$$

where $reading_x$ is the projected reading, $r_{reading}$ is the raw readings of the LiDAR and $theta_r$ is the angle from the IMU.

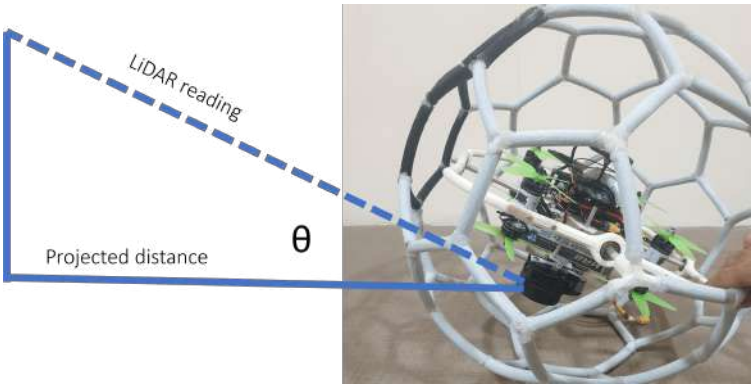


Figure 48. LiDAR reading projection

To further correct the LiDAR readings, a filter was set so that it will ignore readings in a specific range. This was done to avoid two kinds of readings:

- Readings of the cage itself
- Bad readings that were not successfully read (infinite)

Figure 49 compares the raw readings of the LiDAR with the corrected ones when the vehicle is run in one direction. This graph ignores bad readings (Infinite values read).

5.4. Ground truth - Optitrack

An Optitrack with 6 cameras was used to compare the ground truth with actual readings from the drone. As described in Section 2.6, these cameras were first calibrated with

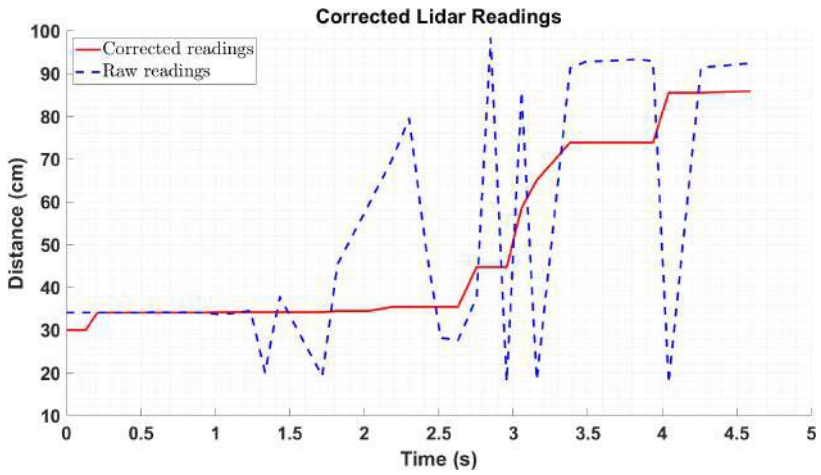


Figure 49. Comparing LiDAR raw readings with corrected projection

known passive markers. The calibration along with the capture volume is shown in Figure 51. A setup for the drone to run was also made. The origin is then marked with the calibration as shown in Figure 50. The start and stop positions of the run inside the optitrack is shown in Figure 52

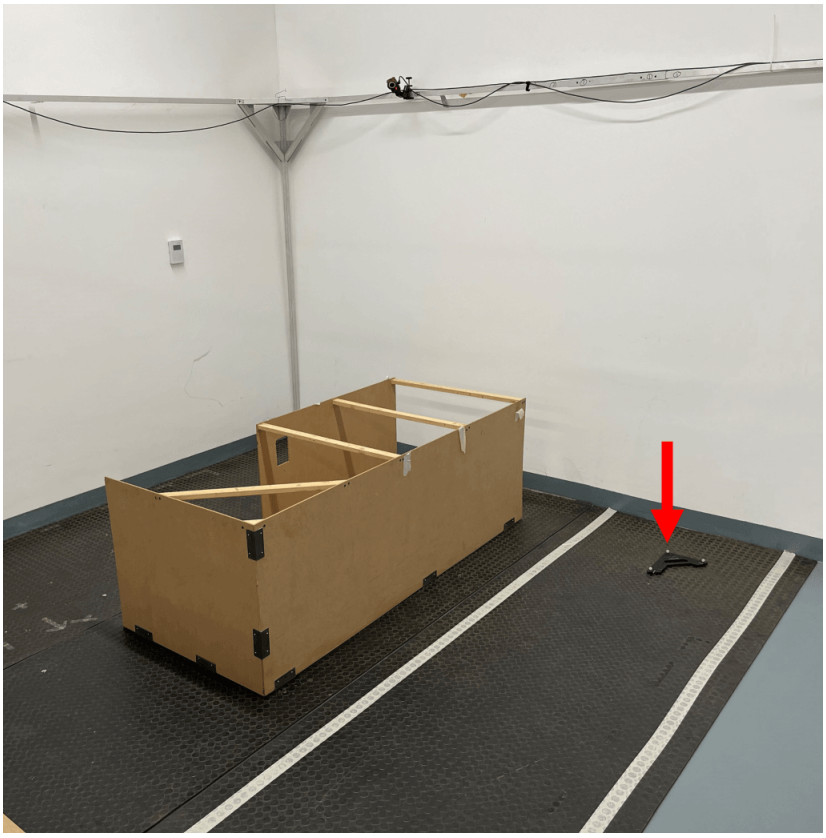


Figure 50. Origin point on drone setup (Duct for comparison)

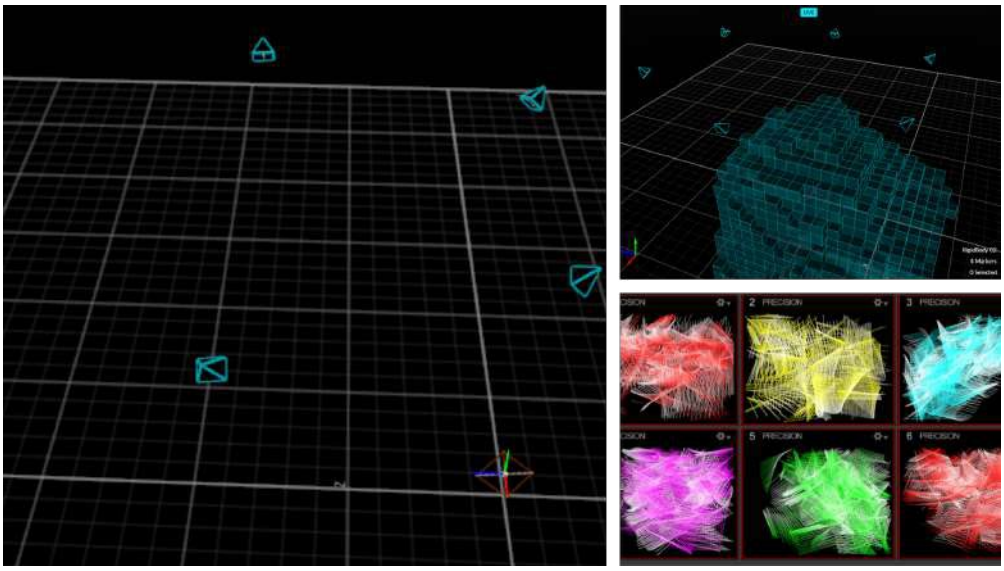


Figure 51. Capture volume and camera calibration

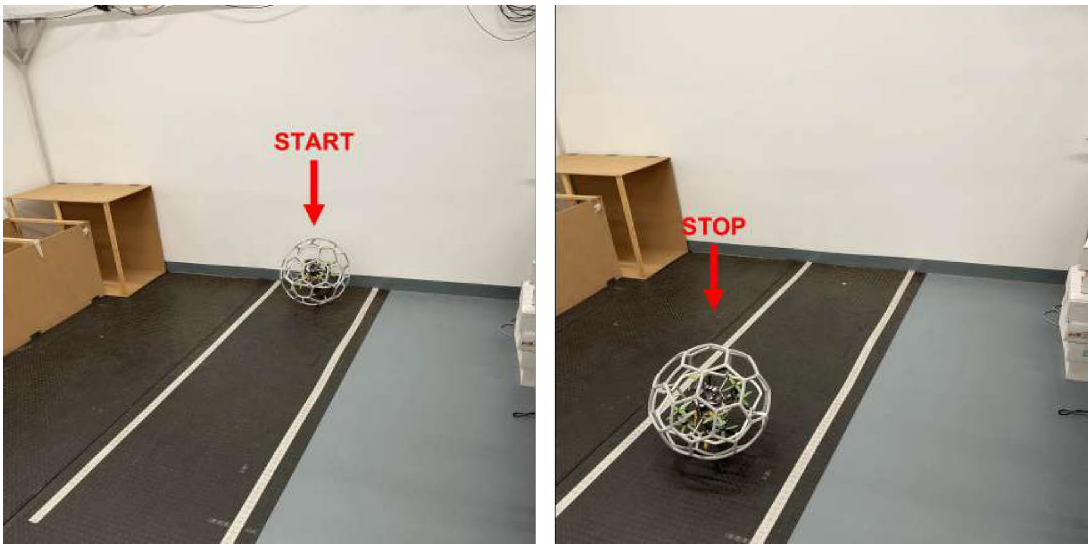


Figure 52. Start and stop position inside Optitrack

Four passive markers were placed on the drone for the cameras to detect. These can be seen in Figure 53. All markers have to be placed such that they do not move relative to each other. This allows us to mark the drone as a rigid body and its position can easily be measured from the cameras. Even if any of the markers are not visible to the cameras at a certain point, the program predicts the unknown marker's position as a rigid body is defined.

The distance measured from the Optitrack and the Lidar was then compared with each other. This is shown in Figure 54.

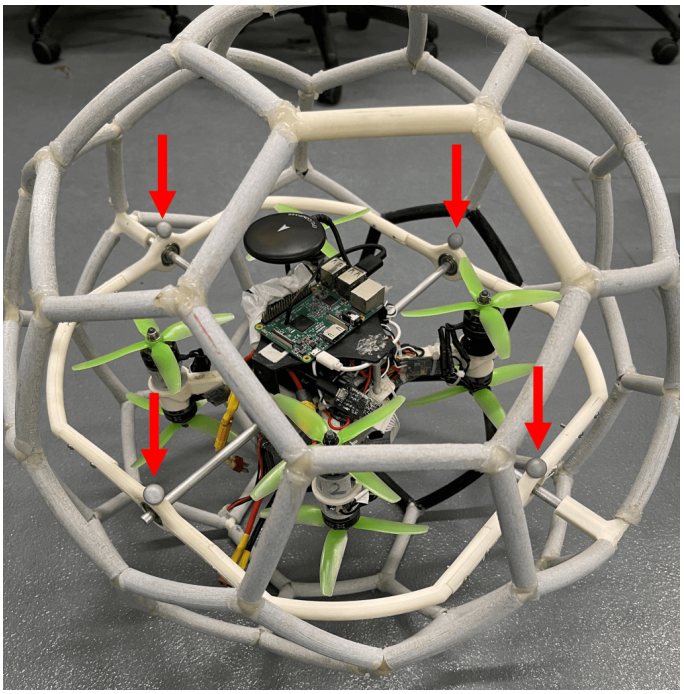


Figure 53. Drone with tracking markers

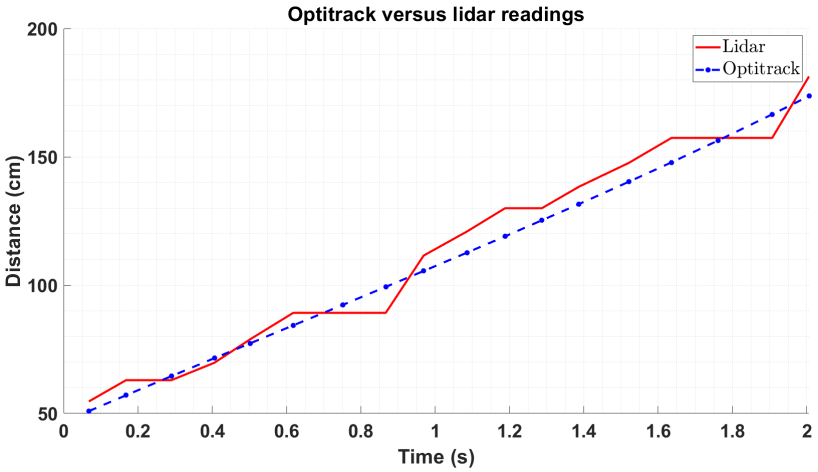


Figure 54. Comparing Lidar data with Optitrack

Error histogram was made from the acquired data and the mean and variance was plotted as shown in Figure 55. The mean and variance are as follows:

$$\mu = 2.924, \sigma = 6.045 \tag{54}$$

The corrected reading can then be written as:

$$\bar{x} = x \pm \bar{x_e} = x \pm \mu = x \pm 2.924 \tag{55}$$

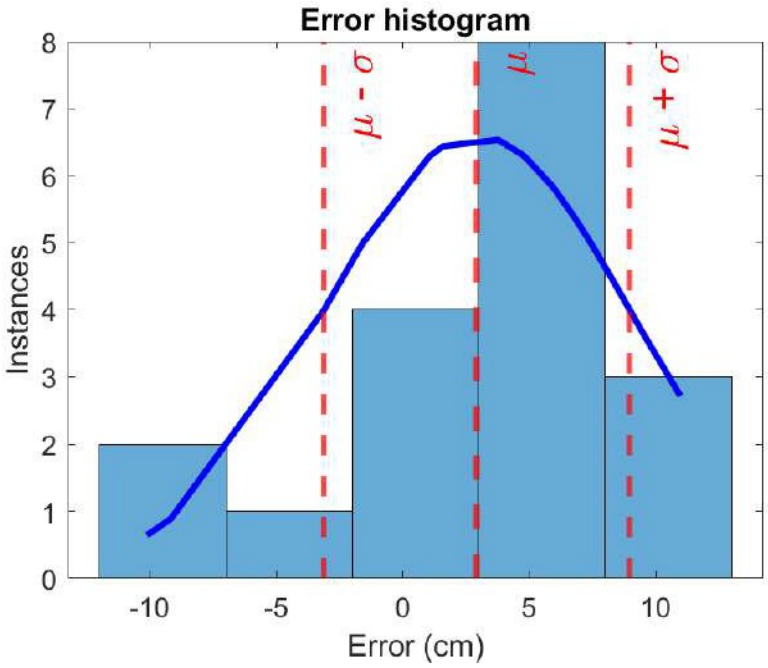


Figure 55. Error histogram with mean and variance

5.5. Running PID and DMRAC on the Vehicle

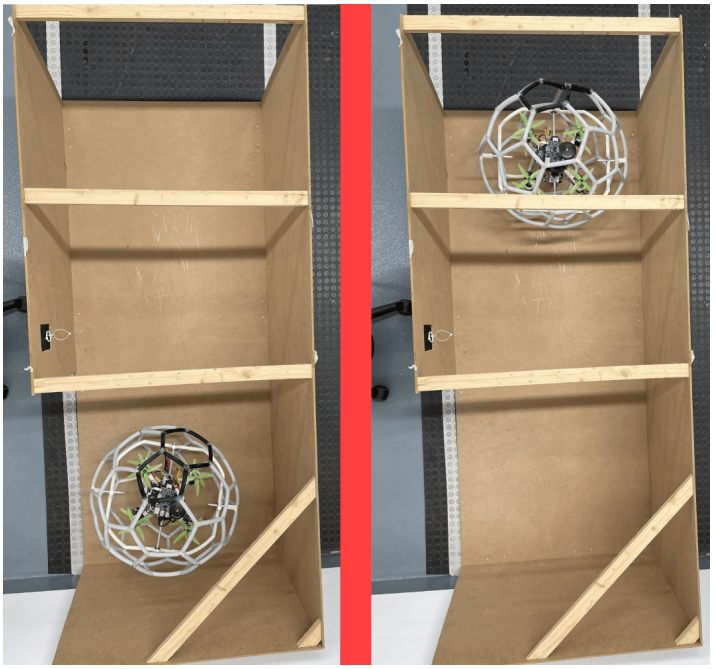
The experiments were run inside a setup that would resemble an industrial HVAC duct. This is done to amplify the fact that the drone can be run in tight spaces without any risk of damaging the moving parts (or the environment around it). Two setups were made for the experiments as shown in Figure 58. Setup 1 was used for autonomous runs that allows motion in one direction only. Setup 2 include motion in another axis which is later used for SLAM. Figure 56 and 57 shows the vehicle moving inside the experimental setup.

5.5.1 PID settings

As explained in Section 5.2, the control law output is converted to PWM and is fed to the theta channel from ROS to the Pixhawk that resembles the angle required to achieve a translation in the 'x' direction. The output to the motors also had hard-limiters as maximum and minimum values to avoid any irregular reading. The PID values were tuned in order to give a stable response from the system. The time step was also set as



Figure 56. Drone inside an industrial HVAC duct. Motion from left to right


Figure 57. Motion from left to right - Top View

constant as follows:

$$K_p = 0.0037, K_I = 0.00001, K_d = 0.003, T_s = 0.06 \tag{56}$$

Increasing K_p resulted in increased instability of the drone. In some cases, the output from the control law blows to infinite because of high K_p value resulting in the drone continuing to move forward indefinitely. Decreasing K_p would result in the drone not producing enough thrust for it to move forward and the vehicle would just sit in the starting position.

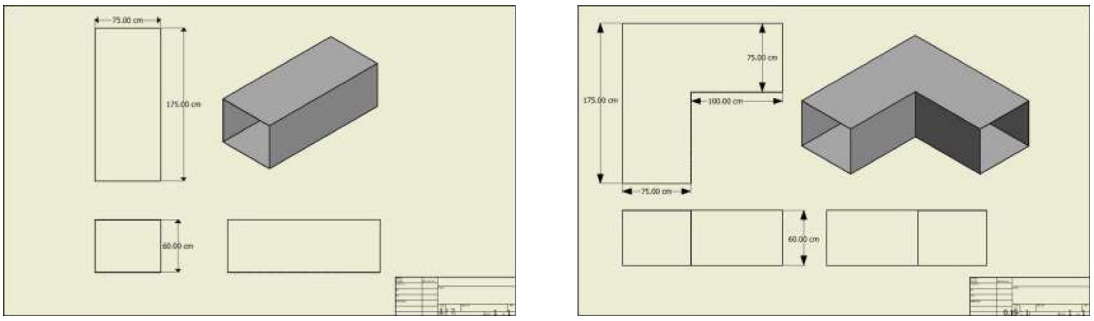


Figure 58. Left: Setup 1. Right: Setup 2

The K_I value was found to be extremely small as increasing it highly increased the instability. This would cause sudden changes in the PWM output, making the vehicle move erratically. Decreasing the K_I (or setting it to zero completely) made the drone react slower and took a long time for it to even move from the starting position.

The K_d value seemed to be the most sensitive of the three. Increasing it caused the drone to move in the opposite direction as gain contributed negatively to the overall sum in equation 12. Decreasing it would almost always cause the vehicle to go unstable run forward indefinitely.

5.5.2 DMRAC settings

Similar to the simulations, the Direct Model Reference Adaptive Control had a PID in conjunction with the adaptive equations. The values for the PID were set the same as they gave stable response. All modifications to avoid the gain drifts were used. However, only the projection modifier (Section 4.3.1.5) gave stable results. The constants used for the model as well as the initial values were the same as the ones used in the simulation (Equations 20 & 46). The gains for the PID were also the same as the gains used in the previous section (equation 56).

5.5.3 Comparing the two control laws

The best runs of both the control laws were studied together and the result is shown in Figure 59.

Figure 59 shows that the adaptive control makes the vehicle to reach its destination faster than the PID control. The difference in the final position and the set-point

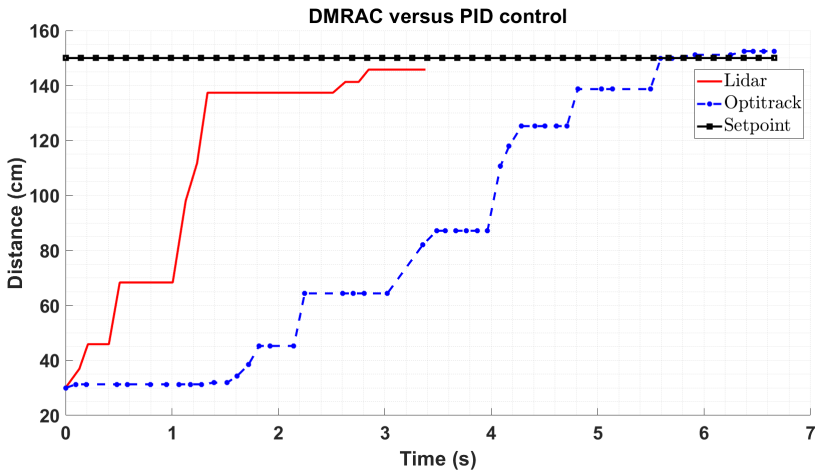


Figure 59. Comparing Adaptive control with PID control

is less than 10 cm which is not significant when compared to the size of the drone. Another observation that was made on the hardware is that the PID control made the vehicle to move slightly in the 'y' axis (sometimes causing it to hit the wall); whereas, the DMRAC did not show thisbehaviour.

5.6. Detecting Thermal Leaks

The next step is to use the thermal camera mounted on the drone to detect thermal leaks. The camera was mounted sideways such that it scans the side of the duct. Figure 60 shows the camera on the drone.

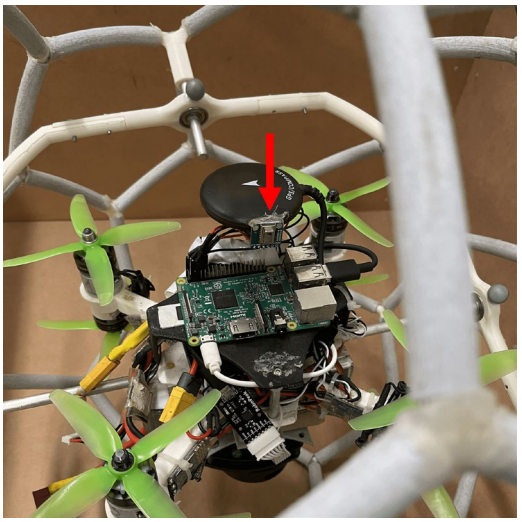


Figure 60. AMG8833 thermal camera on the drone

Bicubic interpolation is done on the scanned image to make it smoother. Figure 61 shows an example of how a scanned image looks like. The color gradient allows us to detect temperature changes. This can be used to detect leaks as well. If there is a leak in an HVAC duct, the temperature around the leak will be different compared to the rest of the duct as hot air will be escaping/entering the duct (depending if the duct is for heating or cooling). This can be seen in Figure 62. To emulate this leak, a hot air blower is used on the side of the duct. This helps create a 'leak' on a known place of '90cm' from the starting of the duct (Figure 63).

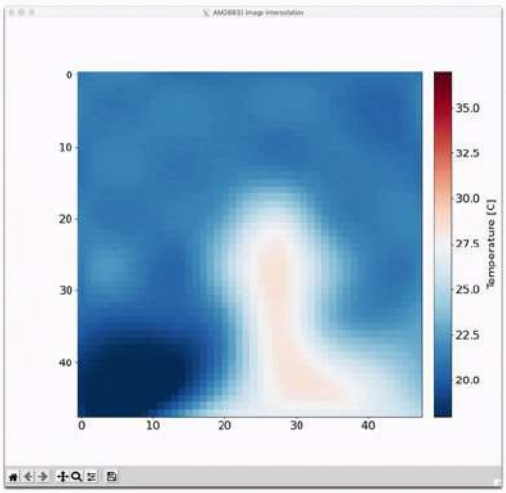


Figure 61. Scanned thermal image

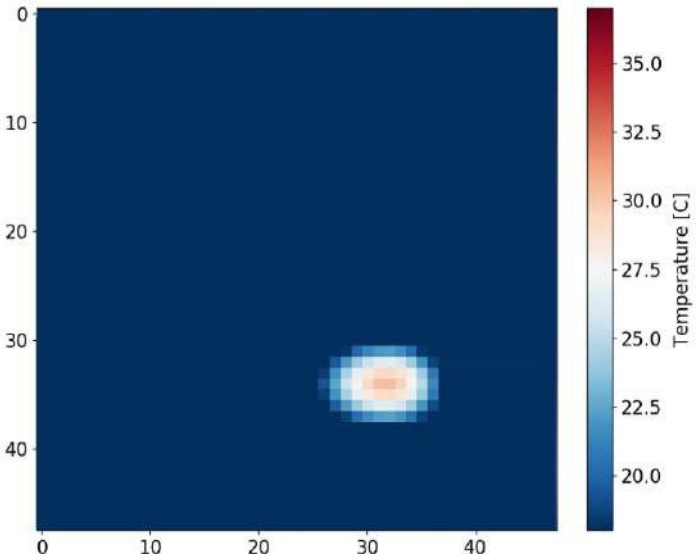


Figure 62. Scanned image of point leak on a duct



Figure 63. Thermal leak from hot air blower

The drone is then run autonomously and the average of the temperature detected is plotted against the distance travelled by the drone. This result is shown in Figure 64.

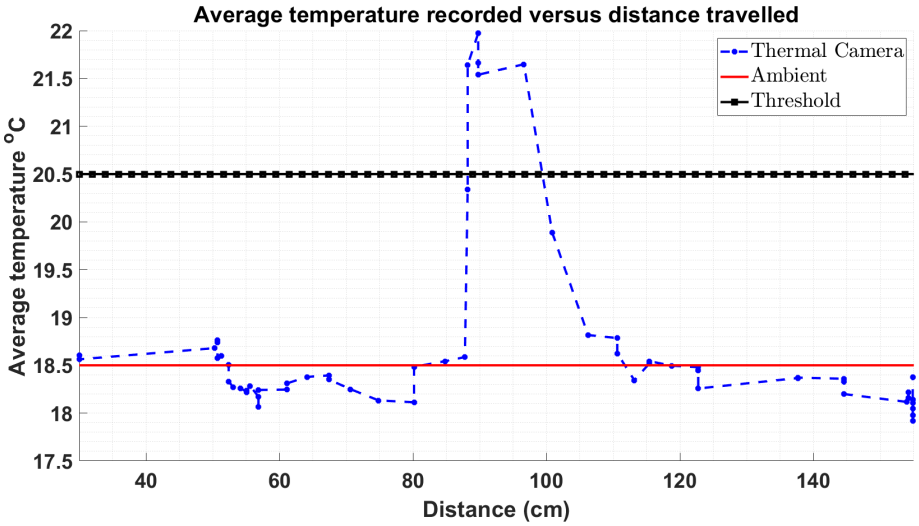


Figure 64. Average temperature detected against distance travelled

As seen from the run, the average temperature rises around 3 degrees due to the air leak. This increase starts from 87.7 cm to 96.6 cm; matching the actual leak distance on the 90cm mark. The threshold can be set to be 2 degrees above the average detected ambient temperature as shown in the Figure.

Chapter 6.2D Localization with Google Cartographer

Having a 2D LiDAR on the drone helps open the possibility of it being able to do simultaneous localization and mapping (SLAM). Since the main frame of the vehicle does not stay parallel to the ground, the pose of the vehicle would also be needed so that the readings from the LiDAR could be corrected. This can be done by using the IMU. A technique for SLAM is required that uses both of these sensors and help generate a map of the surroundings. Google Cartographer [61] is one approach that is based on 2D portable laser-range finders. It generates a real-time map of the surroundings in the 2D plane and loop closure (at 5cm resolution) can also be achieved. An overview of how the approach works is discussed in this chapter. This is then followed by some of the settings in the algorithm specific to our system. Results in the 2D plane are then discussed in the end. If control can be achieved in the 3D space, then SLAM can also be extended in the 3D space.

6.1. Google Cartographer Overview

Google Cartographer [61] uses a branch to bound approach and pose optimization rather than particle filter and graph-based theory. It can accommodate 2D lasers scans, IMUs and odometry readings to give a map of the environment as well as the path taken by the vehicle. Figure 65 shows the general structure of google cartographer.

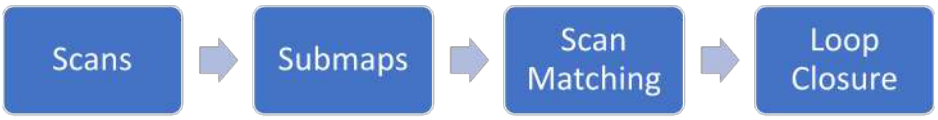


Figure 65. Google Cartographer overview

6.1.1 Scans

The first objective is to get appropriate scans of the surroundings. For an unstable platform, an IMU is used to estimate the orientation of gravity. This helps in projecting the

LiDAR scans to a 2D environment. These scans are then aligned to construct submaps and the transformation matrix that transforms from the scan frame to the submap frame is defined as:

$$T_\xi p = \begin{bmatrix} \cos\xi_\theta & -\sin\xi_\theta \\ \sin\xi_\theta & \cos\xi_\theta \end{bmatrix}_{R_\xi} p + \begin{bmatrix} \xi_x \\ \xi_y \end{bmatrix}_{t_\xi} \tag{57}$$

6.1.2 Submaps

A submap is made up of few successful scans. They take the for of probability grid with discrete grid points ($M : r\mathbb{Z} \times r\mathbb{Z} \rightarrow [p_{\min}, p_{\max}]$). Each point is given a probability value which corresponds to the probability of that particular pixel to be occupied. These probabilities are then updated as more readings are fed to the system through equations 58 and 59. When a scan is made, each reading is classified as a hit or a miss. For every miss in the LiDAR reading, all grid points are added to the miss section that are between the scanned point and the origin. This does not include the already hit grid point.

$$\text{odds}(p) = \frac{p}{1-p} \tag{58}$$

$$M_{\text{new}}(x) = \text{clamp}\left(\text{odds}^{-1}\left(\text{odds}\left(M_{\text{old}}(x)\right) \cdot \text{odds}\left(p_{\text{hit}}\right)\right)\right) \tag{59}$$

6.1.3 Scan matching

Ceres-based scan matcher [62] is used to find a scan pose that maximizes the probability of the scan points in the submap. This can be written as a non-linear least square problem as:

$$\underset{\xi}{\text{argmin}} \sum_{k=}^{K} \left(1 - M_{\text{smooth}}\left(T_\xi h_k\right)^2\right. \tag{60}$$

where T_ξ is the transformation matrix from the scan frame to the submap frame based on the scan pose. Bicubic interpolation is used in the function $M_{\text{smooth}} : \mathbb{R}^2 \rightarrow \mathbb{R}$ which is the smooth version of the probability values in the local submap.

6.1.4 Loop closure

Since scans are only linked to their related submap, having a large number of submaps stitched together will accumulate error. Hence loop closure optimization is done. This is also formulated as a nonlinear least squares problem

$$\underset{\Xi^m, \Xi^s}{\operatorname{argmin}} \frac{1}{2} \sum_i \rho\left(E^2\left(\xi_i^m, \xi_j^s; \Sigma_{ij}, \xi_{ij}\right)\right) \tag{61}$$

where submap poses are defined by $\Xi^m = \{\xi_i^m\}_{i=1,\ldots,m}$ and the scan poses are $\Xi^s = \{\xi_j^s\}_{j=1,\ldots,n}$. The constraints take the form of relative poses ξ_{ij} and associated covariance matrices Σ_{ij}. The target is the optimal pixel accurate match using brand-and-bound scan matching

$$\xi^* = \underset{\xi}{\operatorname{argmax}} \sum_{k=}^{K} M_{\text{nearest}}\left(T_\xi h_k\right) \tag{62}$$

where W is the search window and M_{nearest} is M extended to all $\mathbb{R}^2$. Using the law of cosines the step-size δ_θ is selected such that the scan points at the max range $d_{\max}$ are not more that r. Therefore:

$$d_{\max} = \max_{=1,..k,\ K} h_k \tag{63}$$

$$\delta_\theta = \arccos\left(1 - \frac{r^2}{2d_{\max}^2}\right) \tag{64}$$

A finite set W is made by using an integral number of steps covering given linear and angular search window sizes. Different algorithms are then presented in [61] to find ξ^* efficiently for large search windows.

6.2. Frames Identification and ROS Settings

To run the cartographer on ROS, frames were first defined to each sensor. This is then linked to the base frame of the vehicle which is then further linked to the map frame. The frame relation can be seen in Figure 66 and the visual frames can be seen in Figure 67.

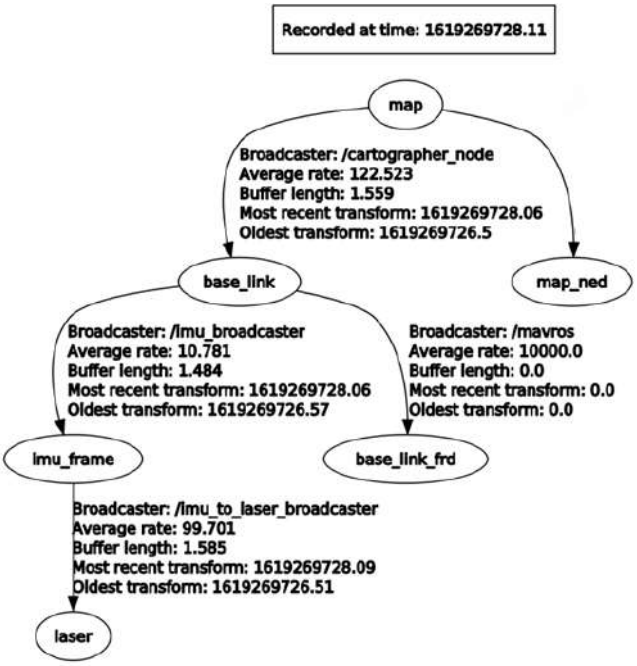


Figure 66. ROS frames definition

Some settings are also done for the configuration of the drone. These are set based on the configuration guide provided by the authors of google cartographer themselves. Both the *tracking_frame* and *published_frame* were set to *base_link* which is the same frame the IMU is attached to. This also coincides with the center of the drone where the Pixhawk lies. The setting for *publish_frame_projected_to_2d* is set to true as we are using the IMU to project the LiDAR readings in the 2D frame. Odometry is not used as we do not have wheels on the vehicle. It would be an interesting topic to see if an encoder could be attached to the shaft of the cage and the rotations be used for the odometry frame. This, however, is not done in this thesis. GPS also is not used to attract applications with GPS-denied areas. Other settings such as the number of laser scans

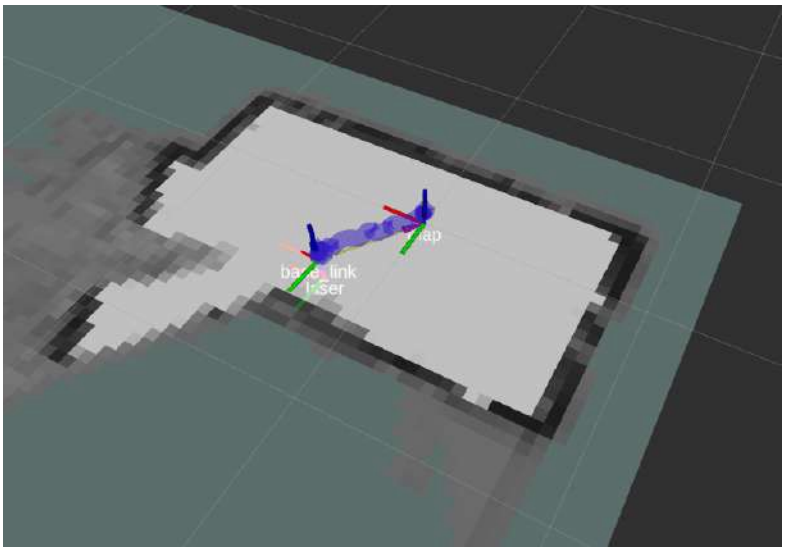


Figure 67. Visualizing the frames

and its range is also set. This is done to ignore the LiDAR readings that are reflected back from inside the cage itself.

The Raspberry Pi was used in a master-slave configuration with a laptop. This was done so that it decreases the work load on the Raspberry Pi and help get readings faster. The cartographer was run on the laptop using the sensor readings published by the companion computer.

6.3. Results

The vehicle was run in a small setup to highlight the vehicle's capability of motion in tight spaces. This can be further extended to have more features or even 3D motion once concrete control is established on it. The drone inside this setup is shown in Figure 68.

The map was generated in ROS 'rviz'. This is shown in Figure 69. Some readings can be seen that were detected by the LiDAR that were away from the actual intended wall. This could be caused when the inner frame is tilted and the LiDAR plane exceeds over the obstacles that needs to be mapped. More refining and tuning is required so that the map ignores such readings and is able to projected the LiDAR readings appropriately. The path that the drone takes is also shown in Figure 69. This shows the trajectory taken by the vehicle to run on the course.



Figure 68. Vehicle in motion for mapping a small area

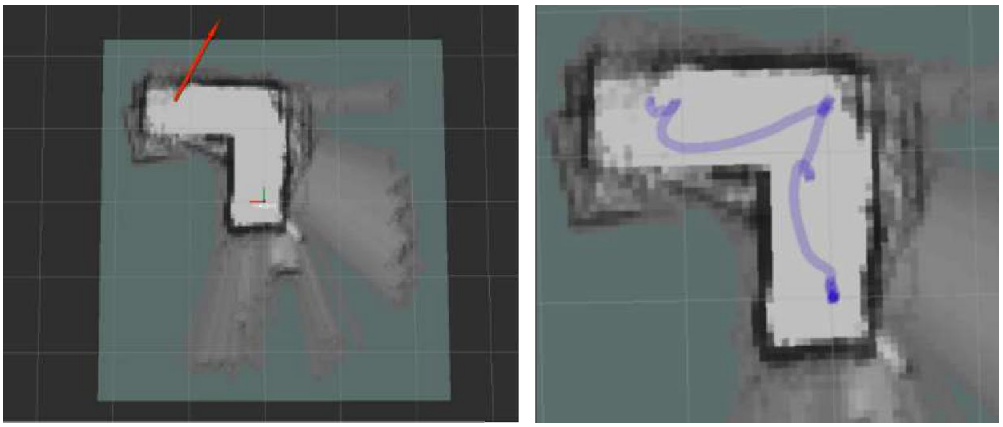


Figure 69. Map generated by Google Cartographer

It is important to mention that the control of the vehicle is done non-autonomously. Future works could include controlling the drone in 3 dimensions and then running the SLAM for a 3D environment. This chapter, however, presents the first steps in using the vehicle to autonomously map a 3D environment.

Chapter 7.Concluding Remarks and Future works

In this thesis, we design a fixed rotor drone with a passive protective cage. Its main usage is in compact environments with good control handling techniques. Simulation is done on a quadmodel using Direct Model Reference Adaptive Control with matched non-linearities. Various modifications are applied to it in order to avoid gain drifts. It is seen that dead-zone modification lowers the rate of the drift compared to no modifications. The sigma and epsilon modifications have similar gain response. The projection based modification clips the gains at a certain predefined value. Increasing the ε value in the dead-zone modification decreases the gains but increases the error between the actual and the model reference. Increasing the damping coefficient in the sigma modification decreases the gains and the error. Increasing the norm-bound in the projection modification increases the gains but decreases the error.

This is then followed by 3D printing the design made and experimenting on an actual platform. The vehicle has a 2D LiDAR that scans the area around the drone in a 2D plane. Since the main frame of the vehicle is able to rotate independent of the cage, an IMU is used to project the readings from the LiDAR to the horizontal plane parallel to the ground. Autonomous control is done in compact space using PID and DMRAC. The results show that DMRAC helps the vehicle achieve its destination in less time compared to the PID control algorithm. The DMRAC also does not make the drone tilt to another axis and hence avoids the vehicle hitting the wall.

The distance measured from the LiDAR is then compared with the Optitrack using passive markers on the drone. This is then used to generate an error histogram and the mean error and variance is plotted. Following this, the drone is run autonomously and using the thermal camera, an average of the temperature detected along a wall is detected. This average temperature is plotted against the distance travelled and the leak is localized based on the peak generated on the plot.

Using the existing sensors, SLAM is also done using Google Cartographer. A small map is made of the surroundings as the drone is ran on an experimental setup.

Further research and improvements on the vehicle can include the following:

- For the hardware, the design could be printed by a lighter material in order to improve the drone properties. This could include expensive material such as carbon

fibre. More flexible materials can also be researched on (such as eLastic TPE) so that the cage can withstand higher loads upon an impact. Structurally, an additional gyroscopic ring can also be added to help have better control in both axes on the ground.

- A control loop to achieve yaw control in the actual setup can be made. This is to ensure that the axis of the drone is always aligned with the duct axis and the drone avoids hitting the sides of the duct. This will also give flexibility in keeping the starting position of the drone itself as it will always fix the yaw position such that it can move parallel to the duct.
- Adaptive control in all three axes can also be researched and then experimented on the vehicle.
- SLAM could further be validated on a larger experimental setup. This could also include mapping in the third dimension.
- After autonomous control is achieved in all three axes, SLAM could then be tried autonomously.

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Appendix A. Derivation of the Quadmodel

A.1. Kinematics of a 6-DOF body

To define the kinematics of any 6-DOF rigid body, two frames are usually defined:

- Earth inertial reference - E-frame
- Body-fixed reference - B-frame

The E-frame is used to define the linear position ($\boldsymbol{\Gamma}^{E}\,[m]$) and the angular position ($\boldsymbol{\Theta}^{E}\,[rad]$. They are also used to describe the difference between each frame. These positions are defined as:

$$\boldsymbol{\Gamma}^{\boldsymbol{E}} = [\,X \quad Y\,Z\,]^{T} \tag{65}$$

$$\boldsymbol{\Theta}^{\boldsymbol{E}} = [\,\Phi \quad \theta\,\psi\,]^{T} \tag{66}$$

The B-frame is used to define the linear velocity ($\boldsymbol{V}^{B}\,[ms^{-1}]$), the angular velocity ($\boldsymbol{\omega}^{B}\,rads^{-1}$), the forces ($\boldsymbol{F}^{B}[N]$) and the torques ($\boldsymbol{\tau}^{B}[Nm]$). The velocities in the B-frame can also be defined as:

$$\boldsymbol{V}^{\boldsymbol{B}} = [\,u \quad v \quad w\,]^{T} \tag{67}$$

$$\boldsymbol{\omega}^{\boldsymbol{B}} = [\,p \quad q \quad r\,]^{T} \tag{68}$$

It is further possible to combine the two vectors in each frame to give a more generalized representation of the rigid body. The E-frame will then define the generalized position $\boldsymbol{\xi}$ and the B-frame will define the generalized velocity $\boldsymbol{v}$.

$$\xi = [\boldsymbol{\Gamma}^{\boldsymbol{E}} \quad \boldsymbol{\Theta}^{\boldsymbol{E}}]^{T} = [\,X \quad Y \quad Z \quad \Phi \quad \theta \quad \psi\,]^{T} \tag{69}$$

$$v = [\boldsymbol{V}^{\boldsymbol{B}} \quad \boldsymbol{\omega}^{\boldsymbol{B}}]^{T} = [\,u \quad v \quad w \quad p \quad q \quad r\,]^{T} \tag{70}$$

To find the relation between the linear velocity in the body frame $\boldsymbol{V}^{\boldsymbol{B}}$ and the linear velocity in the Earth frame $\boldsymbol{\Gamma}^{\boldsymbol{E}}$, a rotation matrix $\boldsymbol{R}_{\boldsymbol{\theta}}$ can be used.

$$\boldsymbol{V}^{\boldsymbol{E}} = \dot{\boldsymbol{\Gamma}}^{\boldsymbol{E}} = \boldsymbol{R}_{\boldsymbol{\theta}}\boldsymbol{V}^{\boldsymbol{B}} \tag{71}$$

This rotation matrix can be identified by using the basic rotation matrices. Using the fixes frame, pre-multiplication of rotation matrices of 'X', 'Y' and 'Z' axes at angles φ . θ and ψ respectively such that:

$$\boldsymbol{R}_{\Theta} = \boldsymbol{R}(\psi, z)\boldsymbol{R}(\theta, y)\boldsymbol{R}(\varphi, x) \tag{72}$$

The basic rotation matrices can be defined as:

$$\boldsymbol{R}(\varphi, x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_{\varphi} & -s_{\varphi} \\ 0 & s_{\varphi} & c_{\varphi} \end{bmatrix} \tag{73}$$

$$\boldsymbol{R}(\theta, y) = \begin{bmatrix} c_{\theta} & 0 & s_{\theta} \\ 0 & 1 & 0 \\ -s_{\theta} & 0 & c_{\theta} \end{bmatrix} \tag{74}$$

$$\boldsymbol{R}(\psi, z) = \begin{bmatrix} c_{\psi} & -s_{\psi} & 0 \\ s_{\psi} & c_{\psi} & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{75}$$

Substituting equations 73, 74 and 75 in 72 to give:

$$\boldsymbol{R}_{\Theta} = \begin{bmatrix} c_{\psi}c_{\theta} & -s_{\psi}c_{\varphi} + c_{\psi}s_{\theta}s_{\varphi} & s_{\psi}s_{\varphi} + c_{\psi}s_{\theta}c_{\varphi} \\ s_{\psi}c_{\theta} & c_{\psi}c_{\varphi} + s_{\psi}s_{\theta}s_{\varphi} & -c_{\psi}s_{\varphi} + s_{\psi}s_{\theta}c_{\varphi} \\ -s_{\theta} & c_{\theta}s_{\varphi} & c_{\theta}c_{\varphi} \end{bmatrix} \tag{76}$$

To find the relation between the angular velocity in the Earth frame $\dot{\Theta}^{E}$ and the body frame $\boldsymbol{\omega}^{B}$, a transfer matrix $\boldsymbol{T}_{\Theta}$ can be used such that:

$$\dot{\Theta}^{E} = \boldsymbol{T}_{\Theta}\boldsymbol{\omega}^{B} \tag{77}$$

The inverse of this transfer matrix can be found as follows:

$$\boldsymbol{\omega}^{B} = \boldsymbol{T}_{\Theta}^{-1}\dot{\Theta}^{E} \tag{78}$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} \dot{\varphi} \\ 0 \\ 0 \end{bmatrix} + \boldsymbol{R}(\varphi, x)^{-1} \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + \boldsymbol{R}(\varphi, x)^{-1}\boldsymbol{R}(\theta, y)^{-1} \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} = \boldsymbol{T}^{-1}_{\Theta} \begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \tag{79}$$

$$\boldsymbol{T}^{-1}_{\Theta} = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\varphi & c_\theta s_\varphi \\ 0 & -s_\varphi & c_\theta c_\varphi \end{bmatrix} \tag{80}$$

Inverting this matrix to get:

$$\boldsymbol{T}_{\Theta} = \begin{bmatrix} 1 & s_\varphi t_\theta & c_\varphi t_\theta \\ 0 & c_\varphi & -s_\varphi \\ 0 & s_\varphi / c_\theta & c_\varphi / c_\theta \end{bmatrix} \tag{81}$$

Combining equations 71 and 77 to relate the generalized position in the E-frame $\dot{\boldsymbol{\xi}}$ with the generalized velocity in the B-frame $\boldsymbol{\nu}$ to get:

$$\dot{\boldsymbol{\xi}} = J_{\Theta}\boldsymbol{\nu} \tag{82}$$

where:

$$J_{\Theta} = \begin{bmatrix} R_{\Theta} & 0_{3\times3} \\ 0_{3\times3} & T_{\Theta} \end{bmatrix} \tag{83}$$

A.2. Dynamics of a 6-DOF body

The effects of forces and torques on a rigid body will be studied in this section. Similar to Section A.1, Newton-Euler approach will be used for the dynamics. The motion equations are easier to write in the body fixed frame as if decreases the complexity for the acceleration terms. Two assumptions are also made:

- The origin of the B-frame is coincident to that of the center of mass (COM) of the body. This will simplify the mass/inertia matrix and hence simplify the overall equations.

- Similar to the first assumption, the axes of the B-frame also coincide with that of the body principle axes of inertia. This will simplify the inertia matrix $\boldsymbol{I}$ to be diagonal.

Writing Newton's second law for the linear components in the E-frame and converting it to body frame:

$$\boldsymbol{F}^E = m\ddot{\Gamma}^E \tag{84}$$

Using the rotation matrix R_Θ:

$$\boldsymbol{F}^E = R_\Theta \boldsymbol{F}^B \tag{85}$$

$$\ddot{\Gamma}^E = \dot{\overline{R_\Theta V^B}} = \dot{R}_\Theta V^B + R_\Theta \dot{V}^B \tag{86}$$

where,

$$\dot{R}_\Theta V^B = R_\Theta \omega^B \times V^B \tag{87}$$

Substituting equation 87 in 86:

$$\ddot{\Gamma}^E = R_\Theta \dot{V}^B + R_\Theta \omega^B \times V^B \tag{88}$$

Using equations 85 and 88 to convert equation 84 to body frame:

$$\begin{aligned} R_\Theta \boldsymbol{F}^B &= mR_\Theta \left(\dot{V}^B + \omega^B \times V^B \right) \\ \boldsymbol{F}^B &= m \left(\dot{V}^B + \omega^B \times V^B \right) \end{aligned} \tag{89}$$

This process can also be repeated for the angular components. Starting with Newton's law:

$$\boldsymbol{\tau}^E = I\ddot{\Theta}^E \tag{90}$$

Using the transfer matrix T_Θ:

$$\boldsymbol{\tau}^E = T_\Theta \boldsymbol{\tau}^B \tag{91}$$

$$\ddot{\Theta}^E = \dot{\underline{T_\Theta \omega^B}} = \dot{T}_\Theta \omega^B + T_\Theta \dot{\omega}^B \tag{92}$$

where:

$$\dot{T}_\Theta \omega^B = T_\Theta \omega^\beta \times \left(I\omega^B \right) \tag{93}$$

Substituting equation 93 in 92 gives:

$$\ddot{\Theta}^E = T_\Theta \dot{\omega}_B + T_\Theta \omega^B \times \omega^B \tag{94}$$

Using equations 91 and 94 to convert equation 90 to B-frame:

$$\tau^B = I\dot{\omega}^B + \omega^B \times (I\omega^B) \tag{95}$$

Combining equations 89 and 95 to get the general motion of a 6-DOF rigid body:

$$\begin{bmatrix} m\boldsymbol{I}_{3\times3} & \boldsymbol{0}_{3\times3} \\ \boldsymbol{0}_{3\times3} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{V}}^B \\ \dot{\boldsymbol{\omega}}^B \end{bmatrix} + \begin{bmatrix} \boldsymbol{\omega}^B \times (m\boldsymbol{V}^B) \\ \boldsymbol{\omega}^B \times (\boldsymbol{I}^B) \end{bmatrix} = \begin{bmatrix} \boldsymbol{F}^B \\ \boldsymbol{\tau}^B \end{bmatrix} \tag{96}$$

where I is the inertial matrix, $I_{3\times3}$ is the identity matrix with dimensions 3 times 3. The mass/inertia matrix in equation 96 is constant and diagonal. This equation is valid for all 6-DOF rigid bodies. However, the last vector (force vector) will be such that it matches the dynamics of a quadrotor. This force vector is evaluated in the next section.

A.3. Deriving the Quadmodel

In this section the quadmodel will be derived. As described in the previous section, the model is very similar to that of a 6-DOF rigid body and its dynamic equation is described in equation 96. This equation can also be written in matrix form as:

$$\boldsymbol{M}_B\dot{\boldsymbol{\nu}} + \boldsymbol{C}_B(\boldsymbol{\nu})\boldsymbol{\nu} = \Lambda \tag{97}$$

where $\boldsymbol{M}_B$ is the system inertia, $\dot{\boldsymbol{\nu}}$ is the generalized acceleration vector, $\boldsymbol{C}_B$ is the Coriolis-centripetal matrix and Λ is the generalized force vector. These can be ex-

pressed as follows:

$$\boldsymbol{M}^{B} = \begin{bmatrix} m\boldsymbol{I}_{3\times 3} & \boldsymbol{0}_{3\times 3} \\ \boldsymbol{0}_{3\times 3} & \boldsymbol{I} \end{bmatrix} = \begin{bmatrix} m & 0 & 0 & 0 & 0 & 0 \\ 0 & m & 0 & 0 & 0 & 0 \\ 0 & 0 & m & 0 & 0 & 0 \\ 0 & 0 & 0 & I_{XX} & 0 & 0 \\ 0 & 0 & 0 & 0 & I_{YY} & 0 \\ 0 & 0 & 0 & 0 & 0 & I_{ZZ} \end{bmatrix} \tag{98}$$

$$C_B(\nu) = \begin{bmatrix} 0_{3\times 3} & -m\,S(V^B) \\ 0_{3\times 3} & -S(I\omega^B) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & mw & -mv \\ 0 & 0 & 0 & -mw & 0 & mu \\ 0 & 0 & 0 & mv & -mu & 0 \\ 0 & 0 & 0 & 0 & I_{ZZ}r & -I_{YY}q \\ 0 & 0 & 0 & -I_{ZZ}r & 0 & I_{XX}p \\ 0 & 0 & 0 & I_{YY}q & -I_{XX}p & 0 \end{bmatrix} \tag{99}$$

$$\Lambda = \begin{bmatrix} \boldsymbol{F}^B & \boldsymbol{\tau}^B \end{bmatrix}^T = \begin{bmatrix} F_x & F_y & F_z & \tau_x & \tau_y & \tau_z \end{bmatrix}^T \tag{100}$$

In equation 99, ${}^{f}S^{f}$ is the general skew symmetric matrix

$$\boldsymbol{S}(\boldsymbol{k}) = -\boldsymbol{S}^T(\boldsymbol{k}) = \begin{bmatrix} 0 & -k_3 & k_1 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix} \qquad \boldsymbol{k} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} \tag{101}$$

The force component, as described in equation 100, can be divided into three components that are directly related to the nature of a quadrotor:

1. Gravitational vector - $\boldsymbol{G}_B(\boldsymbol{\xi})$
2. Gyroscopic effects produced by the propeller rotation - $O_B(\nu)\Omega$
3. Forces and torques produced by the main movement inputs - $\boldsymbol{U}_B(\boldsymbol{\Omega})$

A.3.1 Gravitational force

The Gravitation vector $\boldsymbol{G}_B(\boldsymbol{\xi})$ only affects the linear components of the model as it is a force and not a torque. To incorporate this vector into the model, it has to be represented in the body frame such that:

$$\boldsymbol{F}_G^B = \boldsymbol{R}_\Theta^{-1}\boldsymbol{F}_G^E = \boldsymbol{R}_\Theta^T \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} = \begin{bmatrix} mgs_\theta \\ -mgc_\theta s_\varphi \\ -mgc_\theta s_\varphi \end{bmatrix} \tag{102}$$

Writing it in the generalized force form:

$$\boldsymbol{G}_B(\boldsymbol{\xi}) = \begin{bmatrix} \boldsymbol{F}_G^B \\ \mathbf{0}_{3\times 1} \end{bmatrix} = \begin{bmatrix} mgs_\theta \\ -mgc_\theta s_\varphi \\ -mgc_\theta s_\varphi \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{103}$$

A.3.2 Gyroscopic torque

This contribution to the force vector is due to the imbalance of the rotor speeds as one pair is rotating clockwise and the other is rotating counter-clockwise and the algebraic sum of the rotor speeds will not be zero. Moreover, this will also be a non-zero value if there are different roll and pitch rates. The gyroscopic torque can be written as:

$$\tau_B(\dot{\boldsymbol{\eta}}) = \begin{bmatrix} \mathbf{0}_{3\times 1} \\ -\sum_{k=1}^{4} J_{TP}\left(\boldsymbol{\omega}^B \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right)(-1)^k \Omega_k \end{bmatrix} \tag{104}$$

Since it is a torque it does not effect the linear components of the main equation. ω^B is the rotation rates, J_{TP} is the total rotational moment of inertia around the propeller axis

and Ω_k is the motor speed in *rad/s*. Computing the terms separately:

$$\omega_B \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ p \\ r \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a-0 \\ -p+0 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ 0 \end{bmatrix} \tag{105}$$

For $k = 1, 3$:

$$+ J_{TP} \begin{bmatrix} q \\ 0 \end{bmatrix} \Omega_{1,3} \tag{106}$$

For $k = 2, 4$:

$$- J_{TP} \begin{bmatrix} -p \\ 0 \end{bmatrix} \Omega_{2,4} \tag{107}$$

Summing all terms and writing it in matrix form:

$$J_{TP} \begin{bmatrix} q & -q & q & -q \\ -p & p & -p & p \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \Omega_1 \\ \Omega_2 \\ \Omega_3 \\ \Omega_4 \end{bmatrix} \tag{108}$$

Substituting equation 108 in 104 to get:

$$O_B(v)\Omega = \begin{bmatrix} \mathbf{0}_{3\times 1} \\ -q \\ p \\ 0 \end{bmatrix} \Omega = J_{TP} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ q & -q & q & -q \\ -p & p & -p & p \\ 0 & 0 & 0 & 0 \end{bmatrix} \Omega \tag{109}$$

where

$$\Omega = -\Omega_1 + \Omega_2 - \Omega_3 + \Omega_4, \quad \boldsymbol{\Omega} = \begin{bmatrix} \Omega_1 \\ \Omega_2 \\ \Omega_3 \\ \Omega_4 \end{bmatrix} \tag{110}$$

Ω defines the overall propeller speed and Ω_i defines the speed of individual propeller. Equation 109 shows how only the angular equations affect the gyroscopic effects produced by the propeller rotations.

A.3.3 Forces and torques produced by movement inputs

From aerodynamics, it can be derived that the square of the propeller's speed is proportional to both the forces and the torques. Hence the movement matrix $E_B[+]$ can be multiplied by the Ω^2 to get the movement vector $\boldsymbol{U}_B(\boldsymbol{\Omega})[+]$. Aerodynamic contributions are noted in constants b (thrust) and d (drag) and can be derived separately.

The constant matrix E_B can hence be identified as shown in equation **??**

$$E_B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ b & b & b & b \\ 0 & -bl & 0 & bl \\ -bl & 0 & bl & 0 \\ -d & d & -d & d \end{bmatrix} \tag{111}$$

Where l is the center to center distance and b and d are the aerodynamic constants.

A.4. Finalizing the Quadmodel in Respective Frames

Using the above three contributions mentioned and equation A.3, the quadrotor dynamics can be re-written as:

$$\boldsymbol{M}_B\dot{\boldsymbol{\nu}} + \boldsymbol{C}_B(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{G}_B(\boldsymbol{\xi}) + \boldsymbol{O}_B(\boldsymbol{\nu})\boldsymbol{\Omega} + \boldsymbol{E}_B\boldsymbol{\Omega}^2 \tag{112}$$

To get the generalized velocity vector, the above equation can be written as:

$$\dot{\nu} = M_B^{-1}\left(-C_B(\nu)\nu + G_B(\xi) + O_B(\nu)\Omega + E_B\Omega^2\right) \tag{113}$$

This is defined in the fixed body frame which is helpful for a general 6DOF rigid body. For a quadrotor it is better to have the linear equations with respect to the earth frame and the angular equations with respect to the body frame. This is also known as the hybrid frame of the 'H-frame'. The generalized velocity vector in the H-frame is written as:

$$\zeta = \begin{bmatrix} \dot{\boldsymbol{\xi}}^E & \boldsymbol{\omega}^B \end{bmatrix}^T = \begin{bmatrix} \dot{X} & \dot{Y} & \dot{Z} & p & q & r \end{bmatrix}^T \tag{114}$$

And the dynamics can be re-written in the H-frame as:

$$\boldsymbol{M}_H\dot{\zeta} + \boldsymbol{C}_H(\zeta)\zeta = \boldsymbol{G}_H + \boldsymbol{O}_H(\zeta)\boldsymbol{\Omega} + \boldsymbol{E}_H(\boldsymbol{\xi})\boldsymbol{\Omega}^2 \tag{115}$$

M_H is the same as M_b defined in equation 98. The coriolis-centripetal matrix with respect to the H-frame is:

$$\bar{C}(\zeta) = \begin{bmatrix} 0_{3\times3} & 0_{3\times3} \\ 0_{3\times3} & -S(I\omega^B) \end{bmatrix} \tag{116}$$

The G_H vector becomes:

$$G_H = \begin{bmatrix} \boldsymbol{F}_G^E \\ 0_{3\times1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -m \\ 0 \\ 0 \\ 0 \end{bmatrix} \tag{117}$$

The gyroscopic effects are unchanged as they only affect the angular equations. Therefore:

$$O_H(\zeta)\Omega = O_B(\nu)\Omega \tag{118}$$

To get the movement vector in the H-frame, the linear equations is rotated by the rotation matrix R_Θ such that:

$$E(\xi)\dot{\Omega} = \begin{bmatrix} R_\Theta & 0_{3\times3} \\ 0 & I_{3\times3} \end{bmatrix} \begin{bmatrix} \\ \\ \\ \end{bmatrix} \Omega = \begin{bmatrix} (s_\psi s_\phi + c_\psi s_\theta c_\phi)\, U_1 \\ (-c_\psi s_\phi + s_\psi s_\theta c_\phi)\, U_1 \\ (c_\theta c_\phi)\, U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix} \tag{119}$$

As done before, rearranging the equation to get the derivative of the generalized velocity vector in the H-frame is shown in equation 120

$$\dot{\zeta} = M_H^{-1}(-C_H(\zeta)\zeta + G_H + O_H(\zeta)\Omega + E_H(\xi)\Omega^2 \tag{120}$$

This can hence be expanded in a system of equations as follows:

$$\begin{aligned}
\ddot{X} &= (\sin\psi \sin\phi + \cos\psi \sin\theta \cos\phi)\frac{U_1}{m} \\
\ddot{Y} &= (-\cos\psi \sin\phi + \sin\psi \sin\theta \cos\phi)\frac{U_1}{m} \\
\ddot{Z} &= -g + (\cos\theta \cos\phi)\frac{U_1}{m} \\
\dot{p} &= \frac{I_Y - I_Z}{I_{XX}} qr - \frac{J_{TP}}{I_{XX}} q\Omega + \frac{U_2}{I_{XX}} \\
\dot{q} &= \frac{I_{ZZ} - I_{XX}}{I_{YY}} pr + \frac{J_{TP}}{I_{YY}} p\Omega + \frac{U_3}{I_{YY}} \\
\dot{r} &= \frac{I_Z}{I_{XX} - I_{YY}} pq + \frac{U_4}{I_Z}
\end{aligned} \tag{121}$$

Vita

Ahmed Khalil was born in Dubai, UAE and received his bachelor of science degree with magna cum laude in Mechanical Engineering from American University of Sharjah, UAE in 2018.

Ahmed joined the Mechatronics Graduate Program at AUS in 2018 where he studied and worked as a graduate teaching assistant. His current interests include 3D printing, drones, control systems and mobile robots.